TABLE OF CONTENTS ELECTRICAL

26 00 01 General Electrical Provisions

26 00 20 Electrical Demolition

26 00 26 Submittals and Shop Drawings

26 05 19 Building Wire and Cables

26 05 26 Grounding

26 05 29 Supporting Devices

26 05 33 Raceways and Fittings

26 05 33.16 Outlet, Junction, and Pull Boxes

26 05 53 Electrical Identification

26 05 60 Overcurrent Protective Devices

26 05 83 Wire Connections

26 24 17 Panelboards

26 24 18 SPD (Surge Protection) Equipment

26 27 26 Wiring Devices

26 29 12 Disconnects

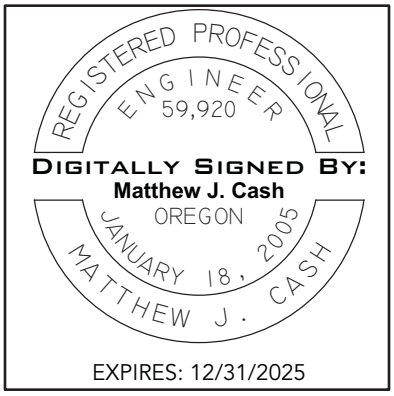
26 29 23 Variable Frequency Drive

26 32 00 Generator Power System (PMG Diesel) (Alternate)

27 05 00 Data System

ELECTRICAL ENGINEER'S SEAL
The undersigned hereby certifies that the Electrical Technical Specifications in this project manual were prepared by me or under my direct supervision.

Matthew Cash, P.E.
Fluent Engineering, Inc.



THE STAMP ON THIS PROJECT APPLIES ONLY TO SECTIONS AUTHORED BY:
MATTHEW J. CASH, PE
FLUENT ENGINEERING, INC.
2110 STATE STREET
SALEM, OR 97301
503-447-5030

SECTION 26 00 01
GENERAL ELECTRICAL PROVISIONS

PART 1 – GENERAL

Products under this contract must meet minimum specifications requirements in detail without exception unless specifically noted and approved as provided in these Specifications. Equipment submitted for review must clearly state on cover sheet any differences from specified product. Equipment substitution or submittal review does not relieve Contractor from meeting all requirements of specified item.

1.1 DEFINITIONS

A. Definitions herein are intended as advisory and shall not limit requirements within the Contract Documents. Where a conflict of definitions exists, the more stringent standard shall be used. Where a term is defined on a Drawing the Drawing definition shall be used for that drawing. Not all definitions are included. Trade standard terms are not defined.

1.2 CONTRACT DOCUMENTS

A. The Contract Documents are inclusive. All requirements of all Contract Documents shall be binding as if repeated herein and within this Division as required by any other Division or Contract Document.

B. This Division does not express or imply separation of the Contract Documents and shall not be considered as separation of the Work.

C. See Advertisement For Bids, Instructions to Bidders, Supplemental Instructions to Bidders, General Conditions, Supplemental General Conditions, Drawings and Specifications, and modifications incorporated in the documents before execution of the Agreement.

D. Conflicts: If any conflicts exist the more stringent is required.

1.3 SCOPE OF WORK

A. General: Provide complete and functional electrical systems as specified, as shown on Drawings, as required, and as intended. Work generally includes, inspections, electrical distribution, lighting, devices, wiring systems, communications systems, raceways, and control systems.

B. Omissions: Contractor shall be responsible for additional labor, or additional material necessary for the proper execution of the Work. Omissions of expressed reference to any item shall not relieve the responsibly to conform to the Contract Documents

C. Scope of Electrical Work

1. All materials and workmanship shall be furnished for complete, tested, and operating electrical systems as shown on the drawings and specified herein.

2. Electrical work is to include the electrical service. Complete to the point of connection with the serving utility. Any changes of or work required by the serving utility, are part of this work and shall be fully included in the bid price.

3. Work is also to include main distribution panel, feeder system and branch circuit panels. Complete branch circuit wiring, switches, receptacles and similar items, exterior lighting and wiring, and wiring and connection to all mechanical equipment, all controls, and civil equipment.

1.4 CONFORMANCE WITH REQUIREMENTS

A. General: All Work shall conform to the reasonable requirements of the project within the scope of the project and authorizations. All work shall conform to the methods and requirements of Code at the location of the Work.

1. Access and inspection: All portions of the Work shall be accessible to inspections and review at all reasonable times during construction. Contractor is responsible for providing access for review and inspection of the Work. Contractor shall secure written inspection reports prior to concealing Work. Contractor is responsible for damages to properly review the Work due to lack of at least 7 Days advance written notification to the Engineer, and Engineer that Work is ready for inspection.

B. Accounting: Provide general accounting information as to labor and equipment costs to assist in determination of modifications to the Contract. Provide accounting breakdown when required for securing Owner financing, or for analysis of equipment costs or equipment payback periods, as well as information for Owner incentives.

1.5 COORDINATION OF TRADES

A. Check all other trade drawings to avert potential installation conflicts. Should major changes from the Drawings be required to resolve potential conflicts, notify the Engineer and secure written approval and agreement on necessary adjustments prior to start of installation.

B. Check all equipment locations and connections on the site for coordination with other Divisions equipment and connections and structure and the like.

C. Contractor is responsible for scheduling trades to properly execute all the Work as intended.

1.6 STANDARD OF CARE AND QUALIFICATIONS

A. General: Contractor shall be experienced and knowledgeable to Provide Work. Owner is not responsible for improper operation, incompliance, or installation due to Contractor's lack of knowledge or experience. Upon request, and where requested herein the Contractor shall supply qualifications and experience. Drawings are presented with industry terms, statements, and trade practices and it is the responsibility of the Contractor to be familiar. Provide written notification prior to Bid to the Engineer if any representation is not understood, or outside standard practice.

B. Like Materials and Quality Control: All systems provided shall be new and of like materials provided through manufacturer authorized distributors. Provide equipment of same system and type by same manufacturer. Items of the same by different manufacturers will be rejected. Equipment shall conform to all applicable Code and applicable listing criteria as of the date of the Contract Documents. Equipment determined to be manufactured under any other listing or Code prior to the date of the Contract is not acceptable, even if the equipment is new or has not been used. All equipment provided to project shall be listed by an approved listing organization.

1.7 EXAMINATION OF SITE

A. Examine Site of Work prior to making Bid. Ascertain all related physical conditions.

B. Verify at the Site of Work prior to Bid scale dimensions shown due to exact locations, distances, and levels will be governed by actual field conditions.

C. Owner will not be responsible for any loss or costs that may be incurred due to a Bidder's failure to fully inform themselves prior to Bid in regard to conditions pertaining to the Work and nature of the Work.

1.8 MINOR DEVIATIONS

A. Make minor changes in equipment locations and equipment connections as directed or required without extra cost.

1.9 RECORD DRAWINGS

A. Maintain a marked set of prints at job site at all times. Show all changes from the original drawing set whether visible or concealed. Include all addendums, field orders, change orders, clarifications, request for information drawn responses, and deviations. Dimension accurately from building lines, floor, or curb elevations. Show exact location, elevation, and size of conduit/raceway, access panels and doors, equipment, and all other information pertinent to the Work.

B. At project completion, submit marked set to Engineer for review.

1.10 TRAINING

A. Provide training of Owner's selected staff for all electrical systems specified herein.

B. Training of all systems shall be videotaped, and two copies shall be distributed to Owner.

C. Notify and Coordinate with Owner for training and attendance not later than 15 Days prior to training.

D. Provide 2 hours of general system training in addition to training indicated below.

1. Domestic Water Control System- 12 hours

2. Variable Frequency Drive- 4 hours

3. Emergency Generator well #3 – 4 hours

4. Emergency Generator well #1 – 4 hours

E. Training shall be conducted by qualified individuals familiar with the Work, and with the equipment.

F. Instructor shall be familiar with programming and operation of equipment and shall provide instruction to do such.

G. Provide contact information to Owner for an additional 1 hours support for all electrical systems.

H. Training shall not occur prior to systems being fully inspected, operational, and complete.

I. Utilize necessary training materials, conduct training at project location including walk-through of equipment on-site.

J. Provide Owner with all required Operation, Maintenance, and Programming manuals provided by equipment manufacturer.

K. Owner shall determine attendees at training, not the contractor. Contractor shall re-train if attendees were not selected by the owner

1.11 **WARRANTY**

A. Warrant Work, materials, and equipment for not less than one year.

B. Provide additional warranty as required herein.

PART 2 - PRODUCTS

THIS PART NOT USED

PART 3 - EXECUTION

THIS PART NOT USED

END OF SECTION

SECTION 26 00 20
ELECTRICAL DEMOLTION

PART 1 - GENERAL

1.1 INTENT

A. The electrical system shall be adjusted to accommodate installation of the new work and is required to meet Code. Necessary adjustments are shown on Drawings and indicated herein.

1.2 EXISTING CONDITIONS

A. The locations of existing utilities and equipment are shown in an approximate way only and have not been independently verified by the Owner or Owner's Representative. Contractor shall determine the exact location of every existing utility before commencing work, and agrees to be fully responsible for any and all damages which might be occasioned by the Contractor's failure to exactly locate and preserve utilities and equipment. Replace damaged items with new material to match existing. Promptly notify Owner if utilities are found which are not shown.

PART 2 - PRODUCTS

THIS PART NOT USED

PART 3 - EXECUTIONS

3.1 DEMOLITION

A. Existing electrical equipment to be removed are denoted on demolition plans. Verify exact location of existing equipment in the field. Only partial existing electrical shown. Locations of items shown on the Drawings as existing are partially based on field inspection of unconcealed equipment and record drawings that may contain errors. The contractor shall verify the accuracy of the information shown prior to bidding and provide such labor and material as is necessary to accomplish the intent of the Contract Documents.

B. Remove existing fixtures, and associated electrical equipment from areas scheduled for remodeling, relocation, or demolition unless specifically shown as retained or relocated on the Drawings.

C. Maintain continuity of existing systems that remain. Remove or relocate electrical boxes, conduit, wiring, equipment, fixtures, etc. as may be encountered in removed or remodeled areas in the existing affected by this work. Wiring which serves usable existing equipment shall be removed and restored clear of the construction or demolition. If existing junction boxes will be made inaccessible, or if abandoned outlets serve as feed through boxes for other existing electrical equipment which is being retained, new conduit and wire shall be provided to bypass the abandoned equipment. If existing conduits pass through areas being removed or remodeled, new conduit and wire shall be provided to reroute clear of the construction or demolition and maintain service to the existing load.

D. Remove all abandoned wiring, and leave site clean.

E. Do not disconnect service without approval from the Owner and Utility. Coordinate demolition work with the utility.

F. Verify with the General Contractor a location for storage of materials, supplies, tools, rubbish, etc. prior to start of Work.

END OF SECTION

SECTION 26 00 26
SUBMITTALS AND SHOP DRAWINGS

PART 1 GENERAL

1.1 REQUIREMENTS

A. Refer to Division 1

B. Organization

1. Refer to Division 1

C. Allow no less than 14 Days for review by Engineer.

D. Contractor is responsible to submit and verify receipt of comments for all submittals.

E. Resubmittals shall contain all items included in previous submittals with changes clearly identified with a cover letter listing the changed items. Only revised items will be reviewed.

F. No item requiring review shall be delivered to the site or otherwise provided to the Project until submittals have been reviewed by the Engineer.

1.2 DEFINITIONS

A. Manufacturing Data: Information regarding the product(s) and equipment issued by the manufacturer as described below.

B. Manufacturer's Label: Manufacturer's label shall include a typewritten list of manufacturer's name, sizes and model or catalog numbers.

C. Manufacturer's Catalog Data: Manufacturer's catalog data shall include standard catalog information (Cut Sheets) marked to indicate specific equipment and options for complete and functional system. All components of the system shall be included. Include listing information. Include installation instructions.

D. Manufacturer's Technical and Engineering Data: Manufacturer's technical and engineering data shall include materials, dimensions, details, installation instructions, weights, capacities, illustrations, wiring diagrams, control diagrams, control schematics, piping diagrams, connection diagrams, performance data, trip curves, listings, mix design, test results, and any other information required for a complete evaluation of the equipment specified, and to verify compliance with the Contract Documents. All available details shall be included with any modifications to the equipment indicated. All manufacturers and associated model numbers used for complete system shall be indicated.

E. Shop Drawings: Shop drawings are Construction drawings of items manufactured specifically for this project. Shop drawings shall include dimensions, construction details, weights, and additional information to identify the physical features of the system or piece of equipment. Drawings shall be adequately sized and scaled for a complete review.

F. Samples: Samples include actual example of the equipment to be installed. Include actual color, finish, and functioning replica of equipment to be installed. Samples will be returned to the Contractor when submitted with pre-paid postage.

G. Certifications and Qualifications: Submit list of past projects (list up to 10) with same systems. Submit information listing references, copies of certificates issued by manufacturer, school, and standards organizations. Submit information mandated in specific specification section.

1.3 SUBMITTALS REQUIRED

A. Product Evaluation Data. 5 copies of product literature. The submittal schedule for product evaluation data is as indicated below. Each item requiring a submittal is given the following code:

L	Manufacturer's Label
C	Manufacturer's catalog data (Cuts)
E	Manufacturer's technical and engineering data
S	Shop drawings
SA	Samples
CR	Certifications
Q	Qualifications

1.4 SUBMITTAL SCHEDULE

Division 26 - Electrical

Section 26 05 19- BUILDING WIRE AND CABLES……C
Section 26 05 26- GROUNDING……C
Section 26 05 29- SUPPORTING DEVICES……L
Section 26 05 33- RACEWAYS & FITTINGS ……L
Section 26 05 33.16- OUTLET, JUNCTION, AND PULL BOXES……C
Section 26 05 53- ELECTRICAL IDENTIFICATION……L
Section 26 05 60- OVERCURRENT PROTECTIVE DEVICES……C,E,S
Section 26 05 83- WIRE CONNECTIONS……L
Section 26 24 17- PANELBOARDS……C,E,S
Section 26 24 18- SPD EQUIPMENT……C,E,S
Section 26 27 26- WIRING DEVICES……C
Section 26 29 12- DISCONNECTS ……C,S
Section 26 29 23- VARIABLE FREQUENCY DRIVE……C,E,S
Section 26 32 00- GENERATOR POWER SYSTEM……C,E,S
Section 27 05 00- DATA SYSTEM……C,E,CR

PART 2 - PRODUCTS

THIS PART NOT USED

PART 3 - EXECUTION

THIS PART NOT USED

END OF SECTION

SECTION 26 05 19
BUILDING WIRE & CABLES

PART 1 - GENERAL

1.1 WORK INCLUDED

A. Wires and Cables

1.2 REFERENCE STANDARDS

A. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code

1.3 DELIVERY, STORAGE, AND HANDLING

A. Deliver new wire and new cable to site in new packaging with standard cable coils/reels. Packaging shall clearly show length, wire size, wire/cable type, and manufacturer.

B. Protect products from weather, moisture, and damage.

PART 2 PRODUCTS

2.1 MATERIALS

A. Building Wiring & Insulation: Copper, 98 percent conductivity, stranded. Solid may be used at contractor's option for wire smaller than #8 AWG. 600 volt insulation, Type THHN for dry interior and damp interior locations. Type THW, THWN or XHHW for wet locations, and exterior locations.

B. Conductor cable with conductors smaller than #12 AWG for branch circuits not permitted.

C. Exterior cables exposed to sunlight shall be listed "sunlight resistant."

D. Control panel wiring no smaller than #14 AWG stranded switchboard Type MTW unless otherwise specified on the Drawings or required by system manufacturer.

E. Motor control wires shall be no smaller than #14 AWG.

F. Wire for other areas as shown on the Drawings.

PART 3 EXECUTION

3.1 INSTALLATION

A. Parallel feeders shall have identical conductor length.

B. Use UL listed pulling lubricant for greater than equivalent #4 AWG wire diameter.

C. Use UL listed pulling lubricant for pulls greater than 75 feet.

D. Remove moisture from raceway prior to wire pull.

E. Provide copper grounding conductors. Provide a ground wire through conduits. Utilize the ground wire as the equipment grounding conductor no smaller than #12 AWG otherwise sized as shown and per NEC.

F. Do not splice feeders, or services. Splices only permitted in accessible junction or outlet boxes where circuit routes deviate. Do not splice or tap branch circuits terminating in a single outlet.

G. Color code conductors per NEC to designate neutral, phase, and ground as follows:

CONDUCTOR	120/208
Phase A	Black, or per (E) facility standard
Phase B	Red, or per (E) facility standard
Phase C	Blue, or per (E) facility standard
Neutral	White
Ground	Green
Travelers	Pink, or per (E) facility standard

CONDUCTOR	480/277
Phase A	Brown, or per (E) facility standard
Phase B	Purple, or per (E) facility standard
Phase C	Yellow , or per (E) facility standard
Neutral	White
Ground	Green
Travelers	Pink, or per (E) facility standard

H. Identify facility High-Leg with signage and consistent coloring (Orange) as required by NEC.

I. Wires shall be factory color coded. Coloring shall be integral to the insulation. Plastic tape permitted on #6 AWG and larger where insulation coloring is not available or practical. Apply tape in spiral half-lap over exposed portions of cable at all locations that cable is accessible.

J. All conductors shall be identified with circuit number where conductors are accessible such as at terminals, outlets, switches, circuit breakers, motor control centers, etc. Identify the ends of a given conductor circuit the same.

K. Do not install wires of different voltage systems in same raceway, box, or other enclosure. Control voltage is permitted in same enclosure only where specific equipment is listed for multiple voltage use, and a listed voltage barrier is provided.

L. Radius of cable bends shall not be less than 10 times the outer diameter of the cable.

M. Do not install cable within conduit per NEC.

END OF SECTION

SECTION 26 05 26
GROUNDING

PART 1 - GENERAL

1.1 WORK INCLUDED

A. Electrical systems grounding.

B. Signal systems grounding.

1.2 APPLICABLE STANDARDS

A. Underwriters Laboratories (UL)

1. UL 467 Standard for Grounding and Bonding Equipment

B. Institute of Electrical and Electronic Engineers (IEEE)

1. IEEE 81 Guide for Measuring Earth Resistively, Ground Impedance, and Earth Surface Potentials of a Ground System Part 1: Normal Measurements

2. IEEE 142 Recommended Practice for Grounding of Industrial and Commercial Power Systems

1.3 APPLICABLE REGULATIONS

A. National Fire Protection Association (NFPA)

1. NFPA 70 National Electrical Code (NEC)

B. NEC references below are based on the 2023 edition. Contractor shall meet current NEC requirements.

1.4 TESTS

A. Measure ground grid resistance with earth test megger and provide additional listed and approved earth grounding devices and conductors as required until resistance is 25 ohms or less. Inform engineer in writing, if resistance is greater than 5 ohms.

PART 2 PRODUCTS

2.1 GROUNDING ELECTRODES

A. Encased Electrode: NEC 250.52(A). One, no smaller than #4 bare solid copper conductor. Install in concrete foundation or footing near contact with earth. Connect to steel reinforcing bars, where available, not less than two times.

B. ~~Ground Rods: 3/4" diameter, 8' long, copper, with approved clamp near surface. Listed as~~ ground rod for direct contact with earth.

C. ~~Grounding Electrode: NEC 250.52(A).~~

2.2 GROUNDING CONDUCTORS, AND JUMPERS

A. Size: Per NEC 250.

B. Material: Copper.

C. Protection: Conductors not in raceway or concealed shall be insulated. Provide raceway where shown or required for physical protection.

PART 3 EXECUTION

3.1 INSTALLATION

A. Regardless of TESTS results above, Provide not less than the following:

1. Underground Metal Cold Water Pipe electrode; If underground Metal Cold Water Pipe is not available Provide Ground Ring.

2. Encased Electrode.

3. Two (2) Ground Rods or Ground Rods as shown.

4. Connection to building steel when present.

B. Grounding Electrodes: Bond all electrodes together. Do not provide other type of electrode than shown without written approval. Provide additional quantity of electrodes as required by 1.03 above.

C. Provide access to all grounding electrode conductor connections.

D. Electrically bypass water meters when required by utility with use of full-size bonding jumper rated for location installed and with pipe clamps routed around and clear of meter.

3.2 POWER AND SIGNAL SYSTEM GROUNDING

A. All equipment grounding conductors shall be routed through same equipment conductor raceway from beginning to end (distribution source to load).

B. Metallic raceways are not approved as equipment grounds.

C. Circuit Grounding: Install grounding bushings, studs, and jumpers at distribution centers, pullboxes, motor control centers, panelboards, and junction boxes.

D. Ground Connections: Clean surfaces thoroughly before applying ground lugs or clamps. If surface is coated, the coating must be removed down to the conductive material. After the coating has been removed, apply a listed and approved noncorrosive compound to cleaned surface and connections. Where galvanizing is removed from metal, it shall be re-applied or painted.

E. Bonding Jumpers: Provide with green insulation and size not smaller than per NEC and larger where shown. Connection to neutral only at service neutral bar. Bonding jumpers shall be contiguous without break, joint, or splice.

F. Service Panel:

1. Connect the various feeder green grounding conductors to the ground bus in the enclosure with suitable pressure connectors.

2. Connect the grounding electrode conductor to the ground bus.

3. Connect the neutral to the ground bus as the main bonding jumper.

4. Connect metallic conduits, which terminate without mechanical conductive connection to the enclosure, by grounding bushings and ground wire to the ground bus.

G. Feeders and Branch Circuits: Install green grounding conductors with feeders and branch circuits. Additional locations and systems as shown.

H. Raceway Systems:

1. Ground all metallic enclosed raceway systems.

2. All enclosed raceway connecting to equipment shall contain a grounding conductor.

3. Conduit systems shall contain a grounding conductor.

4. Bond grounding conductor at beginning and end of raceway provided for mechanical protection containing only a grounding conductor.

I. Boxes, Cabinets, Enclosures, and Panelboards:

1. Bond the grounding wires to each pullbox, junction box, outlet box, cabinets, and other enclosures through which the ground conductors pass.

2. Provide lugs in each box and enclosure for ground wire termination.

3. Provide ground bars in panelboards, bolted to the housing, with sufficient lugs for terminating the ground wires.

J. Receptacles - Refer to Section 26 27 26.

K. Ground lighting fixtures to the equipment grounding conductor of the wiring system.

L. Fixed electrical equipment shall have a ground lug installed for termination of the equipment ground conductor.

M. Motors: Install a separate insulated equipment grounding conductor from the equipment ground connection in the motor controller through the raceway and flexible conduit to the ground terminal on the motor housing. Ground motor controller through feeder raceway. No reductions.

N. Variable Frequency (Speed) Drives (VFD/VSD): Additional grounding requirements as shown on Drawings.

O. Control and Signaling Equipment: Ground metallic enclosures and raceways, terminate shields and drain wires to building ground system. Provide additional grounding as required by equipment manufacturer.

3.3 TESTING

A. Test per IEEE 81.

B. Grounding Electrode Conductor:

1. Measure resistance between switchboard ground bus and each grounding electrode, using a Megger and a single length of additional wire.

2. Measure resistance between both ends of the additional wire used.

3. Grounding Electrode Conductor resistance is the difference between 1 and 2.

4. Correct any inadequate connections as indicated.

END OF SECTION

SECTION 26 05 29
SUPPORTING DEVICES

PART 1 - GENERAL

1.1 WORK INCLUDED

A. Raceway Supports.

B. Cable supports.

C. Provide all hardware and materials to support, as required, a complete and congruent raceway system.

1.2 APPLICABLE STANDARDS

A. National Fire Protection Association (NFPA)

1. NFPA 70 National Electrical Code

B. Underwriters Laboratories (UL)

1. UL 2239 Hardware for the Support of Conduit, Tubing, and Cable

C. National Electrical Contractors Association (NECA)

1. ECA 101 Standard for Installing Steel Conduit

PART 2 - PRODUCTS

2.1 RACEWAY SUPPORTS

A. Single Runs: Steel rod hangers, galvanized single hole conduit straps, or ring bolt type hangers with spring clips. Adhesives, tape, staples, or "J-nails" not acceptable.

B. Multiple Runs: Rack with 25 percent spare capacity. Maximum width per manufacturer's recommendations.

C. Vertical Runs: U-channel support with conduit fittings.

D. All hardware such as inserts, straps, bolts, nuts, screws and washers shall be galvanized or plated steel.

E. PVC coated galvanized steel in exterior and wet locations.

F. Channel manufacturers: Kindorf, Unistrut, or approved.

2.2 CABLE SUPPORTS

A. ~~Approved plastic-coated wire-ties.~~

B. Approved PVC coated hangers.

C. Building studs as permitted by Code and specifications.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Supporting devices shall be listed for the location installed. Supports shall be of like material of raceway and be rated for location installed.

B. Layout to maintain headroom, neat mechanical appearance, and to support equipment loads required.

C. Exact location and spacing between supports per manufacturer's recommendations and NEC requirements.

D. Provide adequate spacing to prevent moisture build-up. All runs of conduit shall be arranged so as to be devoid of traps wherever possible.

E. Cable “Sag” greater than 3-Inches from valley to peak of run, not acceptable.

END OF SECTION

SECTION 26 05 33
RACEWAY & FITTINGS

PART 1 - GENERAL

1.1 WORK INCLUDED

A. Conduit, Fittings, and Tubing.

B. Flexible Conduit.

1.2 REFERENCE STANDARDS

A. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code--Chapter 3.

2. Underwriters Laboratories (UL) 6,

3. UL797

4. UL1990

PART 2 - PRODUCTS

2.1 MATERIALS AND COMPONENTS

A. General: No smaller than ¾-inch unless otherwise shown or indicated herein.

B. Conduit and Tubing: Electrical metallic tubing, galvanized rigid steel threaded conduit, Schedule 40 PVC.

C. Flexible Conduit: Flexible plastic jacketed type with liquidtight connectors and steel wrap armor (liquidtight flexible metallic conduit).

D. Fittings:

1. General: Listed and approved for purpose. Water, gas, concrete tight where required.

2. Electrical Metallic Tubing (EMT): Connectors to be steel. All connectors shall have factory insulated throats. Couplers and connectors shall be compression, setscrew type.

3. Galvanized Rigid Steel Conduit (GRC): Threaded. Do not use pressure type. Provide factory insulated throats on bushings.

4. Liquidtight Flexible Metallic Conduit: Continuous copper ground in core; approved watertight.

E. Expansion Joints: Offset or sliding type with bending straps and clamps. Listed for purpose.

F. Entrance Seal: Shall be modular, mechanical type, consisting of inter-locking synthetic rubber links shaped to continuously fill the annular space between the pipe and the wall opening. The

elastomeric element shall be sized and selected per manufacturer's recommendations. Link-Seal or approved.

G. Underground Marking Tape: 6-inches wide, yellow, low density polyethylene 4 mil thickness. Imprinted: "CAUTION: STOP DIGGING - BURIED ELECTRIC LINE BELOW" and current date. Tape for telephone line similar, except green.

2.2 TYPE

A. Utilize GRC in concrete with concrete-tight connectors.

B. Utilize GRC for exterior with watertight connectors.

C. Utilize electrical metallic tubing concealed in finished interior spaces.

D. Utilize electrical metallic tubing exposed in unfinished spaces, where not subject to physical damage.

E. For underground conduit, utilize Schedule 40 PVC or GRC. Provide GRC elbows and GRC risers through penetrations where PVC is used.

F. Utilize surface metal raceways for exposed runs in finished areas. Paint to match wall finish. Use only where shown on Drawings or where approved.

G. Connections to motors, vibrating equipment, and movable equipment shall be with flexible metallic conduit or liquidtight flexible metallic conduit. Use liquidtight type in damp locations. No smaller than 1/2-inch for motor connections. Use 3/8-inch only for light fixture wiring where provided by light fixture manufacturer. Provide sufficient length of flexible conduit to stop vibration into connecting support. Sizes not noted on the Drawings shall be as required by the NEC and no smaller then upstream connection conduit size.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install raceway concealed in all areas where required. Concealment not required in mechanical and electrical rooms, connections to motors, above suspended ceilings, and underfloor spaces.

B. Coordinate installation of conduit in masonry, cabinetry, and building slab work.

C. Underground Raceways: Watertight, including fittings, slope 3 inches per 100 feet downward from building. Install underground marking tape. Bury 6 inches to 8 inches below grade directly above raceway. Seal exterior junction boxes or provide with drainage.

D. Galvanized rigid steel conduit installed in contact with earth shall be wrapped with 2-half laps of 10 mil, all weather, corrosion protection tape.

E. Route all conduits parallel or perpendicular to building lines.

F. Vertical Runs: Straight and plumb.

G. Raceways Running in Groups: Run at same elevation, properly spaced and supported.

H. Install conduit in concrete slab with minimum 2-inch cover. Do not install conduit larger than one inch maximum in concrete slabs unless approved.

I. Do not interfere with placement of concrete re-bar. Place raceway between re-bar layers. Space at least 8-inches on center. Space as far as possible where terminating at same area. Secure raceway, boxes, inserts, etc. by mechanical means prior to pour.

J. Install conduit free with no dents or bruises. Cap ends to prevent entry of foreign materials and moisture.

K. Clean raceway before installation of conductor.

L. Alter conduit routing to avoid obstructions, minimizing crossovers. Avoid use of bends and offsets where possible. Only bend raceway with an approved conduit bending machine or approved hand (hickey) bender.

M. Provide listed expansion complete fittings with grounding jumpers where conduits intersect building expansion joints and for longer runs where conduit expansion may be excessive.

N. Allow minimum of 6 inches clearance at flues, steam pipes, and heat sources.

O. Dissimilar Metals: Avoid contact with pipe or duct runs of other systems.

P. Lengths and Bends: Maximum number of bends in any run shall be the equivalent of four quarter bends (360 degrees total). Maximum length of any run shall be 300 feet, less 50 feet for each equivalent quarter bend. Provide Junction and pull boxes to meet these limits.

Q. Provide entrance seal for all exterior wall, underground, and exterior slab raceway penetrations.

R. All empty raceways shall be provided with pull string or #12 conductor. Provide #12 conductor for exterior empty PVC raceways.

END OF SECTION

SECTION 26 05 33.16
OUTLET, JUNCTION, AND PULLBOXES

PART 1 - GENERAL

1.1 Description of Work

A. Provide electrical boxes and fittings as required for a complete, protected, and operable system.

B. Comply with local Codes and NEC as required for Providing electrical boxes and fittings.

1.2 Reference Standards

A. American National Standards Institute (ANSI).

1. C73 Series Dimensions of Attachment Plugs and Receptacles

B. National Electrical Manufacturers Association (NEMA)

1. OS 1 Sheet-Steel Outlets Boxes, Device Boxes, Covers, and Box Supports

2. FB 1 Fittings, Cast Metal Boxes, and Conduit Bodies for Conduit, Electrical Metallic Tubing, and Cable

1.3 Applicable Regulations

A. American National Standards Institute (ANSI).

1. C2 National Electrical Safety Code (ANSI/IEEE C2)

B. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code.

C. Underwriters' Laboratories (UL).

1. UL50 Cabinets and Boxes (ANSI/UL50).

2. UL514 Outlet Boxes and Fittings (ANSI/UL514).

PART 2 PRODUCTS

2.1 Outlet Boxes:

A. No smaller than 4-inch, 1-1/2-inches deep box. Provide raised covers where required for surface mounted outlets, plaster rings on flush outlets. Provide tile rings where flush outlets installed in tile. Concrete type where installed in concrete.

B. Receptacle Outlets and Flush Switch: 4-inch square box, 1-1/2-inches deep, with single or two-gang plaster ring.

C. Match one piece gang boxes to number of devices, install one device per gang. Devices requiring more than one-gang shall be installed in individual boxes matched to device size. Do not exceed 5-gang configuration per row of devices at same location.

D. Provide galvanized steel interior dry location outlet wiring boxes for emt raceway shaped and sized, to conform to each individual location and installation. Provide with factory knockouts in back and sides, and with threaded holes with screws for securing box covers or devices.

E. Provide outlet box accessories as required. Accessories include mounting brackets, wallboard hangers, extension rings, fixture studs, cable clamps and metal straps for supporting outlet boxes. Choice of accessories is Contractor's option.

F. Outlet Box Covers:

1. Flush Mounting: Bevelled, white nylon plastic, match device installed or full cover where no device installed.

2. Surface Mounting: Bevelled, steel, pressure formed, match device installed or full cover where no device installed.

2.2 **Weatherproof/ Wet Location Outlet Boxes:**

A. Provide corrosion-resistant cast metal weatherproof outlet wiring boxes, shaped and sized, to conform to each individual location and installation. Provide with threaded conduit ends, suitably configured for each application, including face plate gasket and corrosion proof fasteners.

B. Weatherproof boxes shall have smooth sides, gray finish.

C. Boxes used in contact with earth shall be cast iron alloy with gasketed screw cover and water-tight hubs.

D. Weatherproof Plates: Cast metal, gasketed for switches provide spring loaded sealed door(s).

E. Weatherproof Receptacle Outlet Cover: Cast metal, NEMA 3R, In-Use type, with locking tab. Match device configuration. 3 ¼ -inches internal depth. T&B CK series, or approved

2.3 **Weatherproof Junction and Pull Boxes:**

A. Provide galvanized sheet steel junction and pull boxes, with screw-on covers; of the type, shape and size, to suit each respective location and installation; with welded seams and equipped with stainless steel nuts, bolts, screws and washers.

2.4 **Knockout Closures:**

A. Provide punched-steel knockout closures for steel boxes.

2.5 **Pullboxes**

A. Provide sheet metal in interior dry locations for EMT raceway. Provide cast metal in exterior, or damp locations. Type and material shall conform to National Electrical Code, with screw-on cover.

B. Flush Mounted Pullboxes: Provide overlapping covers with flush head screws, finished in light gray enamel.

C. Box volumes shall meet NEC for size and number of entering conduits and cables.

2.6 **Underground Pullboxes**

A. Underground Pull Boxes: Cast concrete with suitable concrete cover to withhold loads in location installed. Provide heavy-duty traffic cover where installed with vehicle traffic. Cover and box shall not deform and be rated for location installed. Provide drainage and no less than 4 feet compacted gravel below installation. Size, and configuration to match installation. Provide where required, and shown on Drawings.

PART 3 - EXECUTION

3.1 **Installation**

A. Match one piece gang boxes to number of devices, install one device per gang. Do not exceed 4-gang configurations per row of devices at same location.

B. Locate outlet boxes flush other than in mechanical rooms, electrical rooms, and above suspended ceilings. Provide insulation behind box to prevent condensation for boxes mounted in exterior walls.

C. Provide insulation behind box for walls with insulation for sound reduction.

D. Coordinate location and mounting heights with built-in units and cabinetry. Outlet mounting height shall be at same level required for equipment served.

E. When mounting receptacle, or voice/voice outlet boxes above bench or counter, mount box to the side (horizontally) for finished receptacle grounding pole at left.

F. Locate pullboxes and junction boxes concealed above suspended ceilings or in electrical rooms, mechanical rooms, or unfinished areas.

G. Support: Provide adequate support of all outlet boxes. Secure boxes independent raceway, by attaching directly to building structure by approved means.

H. Identify each junction and pullbox with system description including branch circuit numbers of enclosed circuits, and voltage.

I. Secure all raceway to entering boxes with approved bushings, and locknuts.

J. Do not mount boxes back-to-back. Boxes on opposite sides of wall shall be separated by at least 3 inches.

K. Maintain sound transmission and fire properties of surface installed. Provide appropriate fire stop and sound stop materials as required to maintain these properties.

L. Provide separate boxes where two voltage systems have equipment at same location. Provide separate boxes for equipment on emergency power system.

END OF SECTION

SECTION 26 05 53
ELECTRICAL IDENTIFICATION

PART 1 - GENERAL

1.1 WORK INCLUDED

A. Permanent Identification of system components.

1.2 APPLICABLE REGULATIONS

A. National Fire Protection Association (NFPA)

1. NFPA 70: National Electrical Code.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Phenolic Nameplate:

1. Three-layer, white front and back with black core.
2. Neatly engraved through outer layer to show white characters on black background.
3. Beveled edges, print lettering.
4. Other colors as specified or shown. Use red for fire alarm, or fire sprinkler only.

B. Stenciling and Silk Screening: Printed lettering with enamel or lacquer paints. Legends contrasting with the background on which applied.

C. Panelboard Directory Card: ~~Fiberboard typed. Laminate or place in protective cover~~.

D. Concealed Box Labels: Permanent black ink such as "Sharpie" pen with neat and legible writing. Red permanent ink for fire alarm.

E. ~~Concealed Conductor Labels: Listed white tape wrapped around individual conductor or cable,~~ with permanent black ink with printed lettering.

F. Permanent Self-Adhesive Labels: Temperature, moisture and UV resistant. Metallic and reflective background with printed contrasting lettering.

G. Exterior Weather-proof Adhesive Label. Temperature, moisture and UV resistant with printed contrasting lettering.

PART 3 – EXECUTION

3.1 EQUIPMENT TO BE IDENTIFIED

A. Motor starters, panels, lighting panels and the disconnecting devices contained therein.

B. Egress/Emergency Lighting.

C. Remote Emergency Inverters.

D. Disconnects.

E. Control panels, starters, pushbutton stations, pilot lights and other control devices.

F. Transformers.

G. Remote control devices.

H. Conductors at both device and terminal strip terminations for control and instrumentation cables and conductors.

I. Other items as specified, required by NEC, or noted on Drawings.

J. Devices in lighting panels and power panels shall be identified on the panelboard directory card.

K. Receptacles and Switches.

L. Junction Boxes.

M. EV Charging Stations.

3.2 PHENOLIC NAMEPLATES

A. Power panels shall be labeled on the door of the interior with a nameplate. Letters for panels shall be printed and no less than 1/2 inch high.

B. Provide nameplate on switchboard

C. Provide nameplates where specified and as shown.

3.3 EXTERIOR ADHESIVE LABELS

A. Each EV Charger head shall be uniquely numbered or identified corresponding with charger controller and Civil labeling requirements.

B. Label value shall correspond to unit programming via Charge Point Network, FuelForce network, or approved.

C. Each EV Charger shall be labeled on the front face of the charging unit. Characters for EV Chargers shall be printed no less 4” high.

D. Submit labeling scheme.

3.4 EXTERIOR WEATHER-PROOF ADHESIVE LABEL

A. Each EV charger shall be labeled on the charging unit with panel number and circuit number. Do not cover existing nameplates/data.

3.5 APPLYING IDENTIFICATION

A. Stenciled letters shall be applied by brush or by spraying.

B. Nameplates shall be attached with either adhesive or screws. If adhesive is used, it shall adequately adhere to the surface installed.

3.6 IDENTIFICATION REQUIREMENT

A. Indicate Voltage for all concealed labels, and for Disconnects, panelboard and switchboard identification.

B. Identification for disconnecting devices contained in panels and motor control centers shall show the equipment name and location by floor, area, and direction to adequately indicate location of load. Do not include Voltage when the Voltage is the same as for the panel or motor control center.

C. Nameplates on disconnect devices located in the area but not part of a panel or motor control center shall have the equipment name, power source identification, and voltage designation. Nameplates for disconnect devices located remotely from the equipment shall also show the equipment location by floor, area, and direction to adequately indicate location of load.

D. All indicators and controls for control panels, starters, and the like shall be labeled, such as (Start, Stop, On, Off, Reset, Fault, etc.).

E. Panelboard directory cards shall list the circuit numbers and show the equipment name and location supplied by the circuits. Equipment locations shall be shown by floor, area and direction, or by room numbers.

F. Device covers to include serving circuit and panelboard name, include at receptacles, switches, outlets, system furniture connections, floor boxes, etc.

G. Apply clear/ translucent adhesive tape with typed black lettering not more than 1/4” in height via use of label maker.

END OF SECTION

SECTION 26 05 60
OVERCURRENT PROTECTIVE DEVICES

PART 1 - GENERAL

1.1 Work Included

A. Circuit Breakers.

B. Fuses.

1.2 Reference Standards

A. American National Standards Institute (ANSI).

1. C37.16 Preferred Ratings, Related Requirements, and Application Recommendations for Low Voltage Power Circuit Breakers and AC Power Circuit Protectors.

2. C37.17 Trip Devices for AC and General-Purpose DC Low-Voltage Power Circuit Breakers.

3. C37.50 Test Procedure for Low-Voltage AC Power Circuit Breakers Used in Enclosures - Test Procedures.

4. C97.1 Low Voltage Cartridge Fuses 600 Volts or Less.

B. Institute of Electrical and Electronic Engineers, Inc. (IEEE).

1. 20-73 Low Voltage AC Power Circuit Breakers Used in Enclosures: ANSI C37.13.

C. National Electrical Manufacturer's Association (NEMA).

1. FU-1 Low Voltage Cartridge Fuses.

1.3 Applicable Regulations

A. Underwriters' Laboratories (UL).

1. UL 489-72 Molded Case Circuit Breakers and Circuit Breaker Enclosures.

2. UL 198 E Class R Fuses.

3. UL 869 Service Disconnects

4. UL 1066 Standard for Low-Voltage AC and DC Power Circuit Breakers Used in Enclosures

B. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code.

1.4 **Quality Control**

A. Breakers shall be selectively coordinated with feeder breakers.

B. Breakers shall be of the same manufacturer as the switchboard and the panelboards.

PART 2 - PRODUCTS

2.1 **Circuit Breakers**

A. Construction

1. Bolt-on connection to bus.

2. Thermal-magnetic, molded case, with inverse time current overload and instantaneous magnetic tripping except as noted in item 9 herein.

3. Quick-make, quick-break, with tripped indication clearly shown by breaker handle taking a position between ON and OFF.

4. Multiple phase breakers shall have a common internal trip. Do not use handle ties between single pole breakers.

5. Breaker shall be switch (T) rated.

6. Where used as service disconnects, breakers shall be listed for use as service entrance equipment and include locking handle.

7. Fully rated at fault current of panel or switchboard.

8. Selectively coordinated by manufacturer with upstream protection device.

2.2 **GFI Branch Circuit Breakers**

A. Meet construction requirements herein.

B. Ground fault protection with integral push-to-test button.

C. Class 1.

D. Adjustable setting pickup from 0.03 to 30 Amps.

E. Adjustable time delay from instantaneous to 2.0 seconds.

2.3 **Testing**

A. By Manufacturer at factory. Timed thermal trip test and timed magnetic trip test.

B. As required by local authority.

C. Submit test results to Engineer upon request.

2.4 **Fuses**

A. Feeder, Branch Circuit and Service Entrance Fuses: 600 amperes and below, UL Class J or RK1 current limiting type, 600 volt 200,000 ampere interrupting capacity.

B. Motor and Inductive Circuit Fuses: UL class RK5 time delay current limiting type, 600 volt, 200,000 ampere interrupting capacity.

C. Control Circuit Fuses: UL Class J or R current, limiting type, 600V.

D. Manufacturer: Bussmann, or approved

PART 3 - EXECUTION

3.1 **Circuit Breaker Installation**

A. Label each breaker located in switchboard or separate enclosure to indicate load served.

B. Adjust settings on breakers to operate properly under actual field conditions and to provide selective system coordination.

C. Torque breakers to bus per manufacturer's requirements and installation procedures.

3.2 **Fuse Installation**

A. Label each switch to indicate type and rating of fuse installed.

B. All fuses shall be selected to provide selective system coordination.

C. Provide the greater of 10%, or not less than 3 spare fuses of each size, and rating used.

END OF SECTION

SECTION 26 05 83
WIRE CONNECTIONS

PART 1 - GENERAL

1.1 Work Included

A. Wires Connectors

1.2 Reference Standards

A. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code.

B. Underwriters' Laboratories, Inc (UL

1. UL 486A through UL 486E

C. American National Standards Institute (ANSI)

1. ANSI/UL 467

PART 2 - PRODUCTS

2.1 Twist-On Connector

A. Pressure-type wound spring twist on connector.

B. Solderless pressure connectors.

C. Shell rating of 105 degrees C.

D. "Push-On" or "punch" type connectors not permitted.

2.2 Compression Adaptor

A. Dual rated for use with both aluminum and copper cable conductors.

B. Diameter and ampacity as current carrying equivalent copper wire.

C. Pre-filled with approved joint compound

D. Connectors shall be clearly marked with Catalog Number, wire size and color-coded die index number.

E. Burndy "Hyplug" type AYP or equal by T&B, or approved.

2.3 Terminal, Crimp Type

A. Flat; fork tongue, or flat circular matched to terminal size.

B. Color coded to wire size.

C. T&B “Sta-Kon”, or approved.

PART 3 - EXECUTION

3.1 **Installation**

A. Provide Twist-On Connectors at taps and splices for conductors no larger than #10 AWG. Provide only in approved junction and outlet boxes.

B. Provide Compression Adapters for terminating a single conductor into mechanical connectors such as a circuit breaker or set screw lugs. Provide only where required for AL/CU transitions or where lugs require adapters.

C. Provide Crimp terminal at all Control voltage terminal blocks, unless otherwise recommended by manufacturer.

D. Do not nick conductor when stripping insulation. No “ringing”

E. Conductor and cable shall not be reduced at the terminal for connections.

F. Connectors shall be approved and listed for the purpose used.

G. Wrap all twist-on connectors with listed tape to maintain equivalent insulation of wire.

H. Remove any obstructions on connection to maintain continuity prior to installation of connectors, such as paint, dirt, and construction materials.

I. Copper conductors can be terminated in approved compression or mechanical connector, including set screws.

J. Provide slack at equipment to allow for a neat termination, access to conductors, and ability to repair or replace equipment.

END OF SECTION

SECTION 26 24 17
PANELBOARDS

PART 1 - GENERAL

1.1 WORK INCLUDED

A. Provide panelboards incorporating equipment of the number, rating and type specified herein and shown in Panel Schedules.

1.2 REFERENCE STANDARDS

A. American National Standards Institute (ANSI).

1. ANSI C37 Standard for Metal-Enclosed Low-Voltage Power Circuit Breaker Switchgear

B. Institute of Electrical and Electronics Engineers (IEEE).

1. Std. 241-74 Electric Systems for Commercial Buildings.

C. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code.

D. Underwriters' Laboratory (UL).

1. UL 67 Standard for Panelboards.

2. UL 869A Reference Standard for Service Equipment

E. National Electrical Contractors Associations (NECA)

1. NECA 407 Standard for Maintaining and Installing Panelboards

F. National Electrical Manufacturer’s Association (NEMA)

1. NEMA PB 1.1 General Instructions for Proper Installation, Operation, and Maintenance of Panelboards Rated 600 Volts or Less.

1.3 QUALITY ASSURANCE

A. Panelboard breakers shall be selectively coordinated with feeder breakers.

B. Acceptable Manufacturers: Eaton/Cutler Hammer, Square D, Siemens (ITE), or approved.

PART 2 – PRODUCTS

2.1 CONSTRUCTION

A. General**:**

1. Mounting hardware shall be captive type, and shall not pose a danger of dropping onto exposed parts if trim is opened, or removed.

2. Provide Arc-Flash Protection label per NEC.

B. Box**:**

1. Galvanized code gauge steel.

2. 20-inch minimum width; 4-inch minimum gutter space on all sides. Larger as required by manufacturer and NEC.

3. Minimum 4 interior mounting studs.

4. Individual knockouts by manufacturer or field-cut per manufacturer requirements. Concentric knockouts are not permitted.

5. All exterior and interior steel surfaces of panelboard door and trim shall be cleaned and finished with industry standard gray baked enamel paint over a rust-inhibiting phosphatized primer coating approved by the paint manufacturer. Panelboards exposed in finished spaces shall have factory finish to match adjacent surfaces.

C. Bussing**:**

1. Copper- All phases, ground, and neutral.

2. Fully Rated. No reductions.

3. Bolt-on breaker Lugs

4. Tap Arrangement shall be phase sequence type, permitting any breaker configuration mounted at any location.

5. Short Circuit Bracing: Fully rated.

6. Neutral Bussing:

   a. Full size, unless otherwise noted.

   b. Properly sized lug for each outgoing neutral connection.

7. All bolts used to connect current-carrying parts together shall be accessible for tightening from the front of the panel.

8. Wiring terminals: Compression or setscrew type for copper conductors; bolted to bus.

9. Secured not less than at 6 separate locations to the enclosure. Bus movement in any direction in excess of 1/8-inch. is not acceptable.

D. Trim**:**

1. Code gauge steel.

2. Flush panels shall include trim overlap around box not more than 1.5-inchs.

3. Surface Panels: Same width and height as box.

4. Mountable by screwdriver, no special tools.

5. Tamper-proof: Trim shall not be removable with door closed.

6. Include separate trim door and access door. Door-in-Door type construction.

7. Doors:

   a. Shall conceal all breakers unless otherwise noted.

   b. Provide internal concealed access door hinges. Exterior door hinge is not required to be concealed, but shall be painted to match exterior color of panel.

   c. Over 48-inches in Height: Shall have auxiliary fasteners at top and bottom of door in addition to flush latch (3-point).

   d. Latches:

      i. Spring-Type, flush, number as required by manufacturer

8. Where more than one latch is installed, access door shall be operable by only one person. Three simultaneously operated latches are not acceptable.

9. Equip latches with flush locks keyed alike. All panelboards shall be keyed alike.

E. Not less than NEMA 1 unless otherwise noted or otherwise required for location installed.

2.2 CIRCUIT BREAKERS

A. Main Breaker:

1. Provide where required and noted. Mount separate from branch breakers.

2. Covered by a metal plate, except for the operating handle, trip flag (if applicable), and trip button (if applicable).

3. Connection to breaker load side shall be contiguous bus bar. Connection by separate conductors is not acceptable.

4. Panel and breaker shall be listed as service disconnect where used in such applications, including alternate power service such as from an emergency generator.

B. Branch Breakers:

1. Bolt-on connection to bus snap-on permitted as per Section 16180. Securely mounted; Deflection of breaker with force applied in any direction greater than ¼-inch is not acceptable.

2. Additional requirements are noted elsewhere in these Specifications, shown on Drawings, and as per Code.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Provide mounting brackets, busbar drillings, and fillers for unused spaces.

B. Maintain fire properties of surface installed. Provide appropriate fire stop materials to maintain these properties.

C. Prepare and affix typed directory to inside cover of panelboard indicating loads and location of loads controlled by each circuit.

D. Provide panelboards flush in all finished areas. Unfinished areas include mechanical rooms and electrical rooms.

E. Securely mount raceway to panelboards with appropriate bushings. Bushings shall be insulated, and threaded through panelboard with locknut securing to box. Coordinate raceway size and locations entering box for neat a professional installation, as well as for appropriate routing.

F. For flush panels, provide (1) 2-inch spare, (2) 1-inch spares, and (2) ¾-inch spares raceways stubbed to accessible ceiling space.

G. Combine all keys on one key ring and furnish to Owner at time of substantial completion.

END OF SECTION

SECTION 26 24 18
SURGE PROTECTION DEVICE (SPD) EQUIPMENT

PART 1 - GENERAL

1.1 DESCRIPTION

A. Provide surge suppression device (SPD) (previously known as TVSS- Transient Voltage Surge Suppression) equipment for the electrical distribution system.

1.2 REFERENCE STANDARDS

A. Underwriters Laboratories (UL)

1. UL 1449 Standard for Transient Voltage Surge Suppressers

2. UL 1283 Standard for Safety for Electromagnetic Interference Filters

B. American National Standards Institute (ANSI)

1. C62.41 IEEE Recommended Practice for Surge Voltages in Low-Voltage AC Power Circuits

PART 2 - PRODUCTS

2.1 INTEGRAL PANELBOARD SURGE SUPPRESSOR

A. Provide in every Panelboard.

B. Unit shall be rated for Category B location per ANSI/IEEE C62.41-1991.

C. Unit shall be factory integrated into panelboard.

D. Each unit module shall be fused with a surge rated fuse and incorporate a thermal cutout device.

E. Minimum surge current capability shall be 80,000 amperes per phase, measured between L-N & L-G.

F. Unit diagnostics, mounted in door of distribution equipment, shall include the following:

1. Operational LEDs to indicate loss of protection and circuit fully operational for each protection status.

G. The internal design of the unit shall have a minimum EMI/RFI filtering of –50 dB from 100kHz to 100MHz

H. The UL 1449 Listed and Recognized Component Suppression Voltage ratings shall not exceed:

1. 330 Volts for 120/208 or 240 voltage configurations.

2. 700 Volts for 480 voltage configurations.

I. All required “burn-in” tests shall be conducted at factory, prior to shipment.

J. Rated for not less than 200,000 AIC.

K. Installation shall not decrease available over-current protection device spaces including main breaker.

L. Acceptable manufacturer: Advanced Protection Technologies, by panelboard manufacturer, or approved.

PART 3 - EXECUTION

3.1 INSTALLATION

A. System shall be complete.

B. Comply with applicable Sections of this Division.

3.2 WARRANTY

A. Full five-year warranty by manufacturer and includes unlimited replacement of all components.

END OF SECTION

SECTION 26 27 26
WIRING DEVICES

PART 1 - GENERAL

1.1 WORK INCLUDED

A. Wall Switches.

B. Receptacles.

C. Ground Fault Receptacles.

1.2 REFERENCE STANDARDS

A. American National Standards Institute (ANSI).

1. C73 Series Dimensions of Attachment Plugs and Receptacles.

B. National Electrical Manufacturer's Association (NEMA).

1. WD 6 Wiring Devices- Dimensional Requirements

2. WD 1 General Color Requirements for Wiring Devices.

C. National Fire Protection Association (NFPA).

1. NFPA 70 National Electrical Code.

D. Underwriters' Laboratory (UL).

1. UL-20 Standard for Snap Switches.

2. UL 498 Attachment Plugs and Receptacles

3. UL 467 Grounding and Bonding Equipment

4. UL 514D Cover Plates for Flush-Mounted Wiring Devices

5. 2006 UL 943 Safety for Ground-Fault Circuit-Interrupters

1.3 QUALITY ASSURANCE

A. Provide type 5362 receptacles.

B. Acceptable Manufacturers: Hubbell, Pass and Seymour, Arrow-Hart, Leviton, or approved

PART 2 - PRODUCTS

2.1 MATERIALS

A. Switches: 120/277 Volt. AC Quiet, slow make, slow break design, toggle style handle, with totally enclosed case, 20 Ampere, specification grade. Provide matching two-pole, three-way and four-way switches.

B. Receptacles: 20 Ampere (unless otherwise indicated), 125 Volts (unless otherwise shown), duplex, polarized, full gang size, specification grade, separate ground terminal, 20 Amp. break-off tab for split circuit wiring.

C. Ground Fault (GFI) Receptacles: 20 Ampere, specification grade duplex receptacle with integral ground fault circuit interrupter. LED operation indicator. Test and reset buttons. End of life protection- GFI component failure results in no power delivered to equipment (2006 UL 943).

D. Wall Plates: 302 Stainless Steel, Match device configuration.

E. Colors:

1. Gray Receptacles, Gray Switches: In all finished areas that are not dark brown in color.

2. Brown Receptacles, Brown Switches & Brown Nylon Wall Plate: In all finished areas with dark brown or dark wood finish.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Do not use back wiring wells, terminate conductors on mechanical screw terminals.

B. Provide wiring devices as shown.

C. Install devices plumb and consistent with building lines. Wall Plates shall make contact on four corners and shall fit flush with device.

D. Devices to include same configuration outlet box, cover, wall plate and other necessary installation materials for a complete operating circuit.

E. Mount switches 42 inches (to center line of faceplate) above floor except as otherwise noted on the Drawings.

F. Coordinate mounting locations with equipment provided.

G. Mount receptacles vertically at 15 inches (to bottom of faceplate) above finished floor, with grounding pole at bottom.

H. Coordinate receptacle height with benches and counters.

I. When mounting receptacle above bench or counter, mount receptacle to the side with grounding pole at left.

J. Grounding: Install a separate bare conductor between the receptacle strap grounding (green) screw and a screw into the outlet box. Self-grounding strap not approved as grounding means.

END OF SECTION

SECTION 26 29 12
DISCONNECTS & MANUAL STARTERS

PART 1 - GENERAL

1.1 WORK INCLUDED

A. Provide motor disconnects as shown, and as required by Codes.

B. Provide circuit disconnects as shown, and as required by Codes.

C. Provide manual motor starters for single phase motors below one horsepower where disconnect is shown as required by Codes.

D. Disconnects to include adequate support, required hardware, and accessories for com-plete functional installation.

1.2 APPLICABLE REGULATION

A. Conform to National Electrical Code and inspection authority.

B. Provide disconnects rated for the location installed, as required by National Electrical Code, as shown, and as indicated herein.

1.3 REFERENCE STANDARDS

A. Underwriters' Laboratory (UL).

1. UL-98 Enclosed Switches.

B. National Electrical Manufacturer's Association (NEMA).

1. NEMA KS-1 Enclosed Switches.

PART 2 - PRODUCTS

2.1 DISCONNECT

A. Motors and circuit disconnects shall have a UL label.

B. Construction: Dry, Indoor Locations shall be not less than NEMA 1. Enclosures for outdoor, or wet locations shall be not less than NEMA 3R. Rated at 600 Volt. Heavy duty, quick make, quick break. Number of poles and ampacity as noted or required by Code. Fusible with dual element fuses where shown. Short circuit rating sufficient to withstand the available fault current. Solid ground, solid neutral.

C. Compression lugs or set-screw lugs approved for use with copper wire.

D. ON/OFF Positions clearly marked.

E. Lockable in "OFF" position.

F. Interlock:

1. Prevents switch from being opened when "ON."

2. Prevents closing switch when cover is open.

3. Provide defeater to permit authorized personnel to open door and inspect switch when "ON," or operate with cover open.

PART 3 - EXECUTION

3.1 INSTALLATION

A. Install motor and circuit disconnects as recommended by manufacturer, required by Code, required by UL, and where shown.

B. Maintain Code clearances and access.

C. Provide a phenolic nameplate on each disconnect identifying the equipment item served.

D. Independently support disconnects. Do not mount disconnects on vibrating, HVAC, or Plumbing equipment.

END OF SECTION

SECTION 26 29 23
VARIABLE-FREQUENCY MOTOR CONTROLLER

PART 1 - GENERAL

1.1 SUMMARY

A. The Variable Frequency Drive (VFD) system shall contain all components required to meet the performance, protection, safety and certification criteria of this specification.

1.2 RELATED SECTIONS

A. Section 26 00 00 – Electrical – General Provisions

1.3 REFERENCES

A. National Fire Protection Association - NFPA 70 - US National Electrical Code.

B. National Electrical Manufacturers Association - NEMA 250 - Enclosures for Electrical Equipment.

C. Underwriters Laboratory Inc. – UL 508.

D. International Electrical Code - IEC 146.

E. Institute of Electrical and Electronics Engineers, Inc. - IEEE 519 - IEEE Standard Practices and Requirements for Harmonic Control in Electrical Power Systems.

1.4 SUBMITTALS

A. Submit under provisions of Section 01 30 00, and Section 26 00 26.

B. Shop Drawings – Approval

1. Elevation Drawings: Include dimensional information and conduit routing locations.

2. Unit Descriptions: Include amperage ratings, enclosure ratings, fault ratings, nameplate information, etc. as required for approval.

3. Wiring Diagrams:

a. Power Diagram: Include amperage ratings, circuit breaker frame sizes, circuit breaker continuous amp ratings, etc. as required for approval.

b. Control Diagram: Include disconnect devices, pilot devices, etc.

4. Major components list.

C. Product Data Sheets

1. VFD and Operator Interface publications.

2. Data sheets and publications on all major components including but not limited to the following:

   a. Contractors

   b. Circuit breaker and fuse (power and control)

   c. Control power transformers

   d. Pilot Devices

   e. Relays/Timers

D. Test procedures shall be per the manufacturer's standards.

1.5 QUALITY ASSURANCE

A. Qualifications:

1. Manufacturers

   a. The VFD and all associated optional equipment shall be UL listed or recognized.

   b. The VFD shall contain a UL label attached on the inside of the enclosure cabinet.

2. Suppliers:

   a. All inspection and testing procedures shall be developed and controlled under the guidelines of the Supplier's quality system and must be registered to ISO 9001 and regularly reviewed and audited by a third-party registrar.

   b. The VFD shall be factory pre-wired, assembled and tested as a complete package.

1.6 DELIVERY, STORAGE, AND HANDLING

A. Contractor shall coordinate the shipping of equipment with the manufacturer.

B. Contractor shall store the equipment in a clean and dry space at an ambient temperature range of -25 °C to 55 °C (-13 °F to 130 °F).

C. The contractor shall protect the units from dirt, water, construction debris and traffic.

1.7 WARRANTY

A. The manufacturer shall provide their standard parts warranty for eighteen (18) months from the date of shipment or twelve (12) months from the date of being energized, whichever occurs first.

B. This warranty applies to variable frequency drive systems.

PART 2 - PRODUCTS

2.1 MANUFACTURERS

A. Allen-Bradley – PowerFlex 755TS VFD with 5% input line reactor and output line filter or approved.

2.2 SPARES

A. Provide one spare units for each well (extra Well #1 VFD, and Extra Well #3 VFD). Prepare for emergency swap out should existing VFD fail. Deliver to location(s) of each as directed by Owner.

2.3 VARIABLE FREQUENCY DRIVE UNIT

A. Features

1. Certifications:

   a. Listed to UL508C

   b. In conformity with EMC Directive (2004/108/EC) and Low Voltage Directive (2006/95/EC). Standards applied; EN 61800-3:2004, EN 61800-5-1:2007

   c. Electric Power Research Institute. Certified compliant with standards SEMI F47 and IEC 61000-4-34

   d. American Bureau of Shipping (ABS)

   e. Lloyd's Register Type Approval certificate

2. Hardware:

   a. Utilize diode bridge on the input rectifier.

   b. Utilize DC bus inductor on all six-pulse VFDs only.

   c. Utilize switching logic power supply operating from the DC bus.

   d. Incorporate phase to phase and phase to ground MOV protection on the AC input line.

   e. Utilize gold plated plug-in connections on printed circuit boards.

   f. Microprocessor based inverter logic shall be isolated from power circuits.

   g. Utilize latest generation IGBT inverter section.

   h. Inverter section shall not require commutation capacitors.

   i. Embedded Ethernet port for direct network cable connections.

j. Battery receptacle for Lithium battery power to the Real Time Clock.

k. Additional DPI port for handheld and remote HIM options.

l. Dedicated Digital Input for hardware enable.

m. Conformal coated printed circuit boards.

n. Optional onboard 24V DC Auxiliary Control Power Supply.

3. Control Logic

a. Ability to operate with motor disconnected.

b. Provide a controlled shut down, when properly protected, with no component failure in the event of an output phase to phase or phase to ground short circuit. Provide annunciation of the fault condition.

c. Provide multiple programmable stop modes including Ramp, Coast, DC-Brake, Ramp-to-Hold, Fast Braking, and Current Limit Stop.

d. Provide multiple acceleration and deceleration rates.

e. Adjustable output frequency up to 650Hz.

f. Ability to control outputs and manage status information locally within the VFD.

g. Ability to function stand-alone or complimentary to supervisory control.

h. Ability to speed reaction time by processing in the VFD.

i. Ability to provide scaling, selector switches, or other data manipulations not already built into the VFD.

j. Ability to read inputs/write outputs and exclusively control the VFD.

k. Ability to provide an option for decision making if communication is lost with main controller.

l. Ability to control other VFDs via a peer-to-peer Ethernet/IP network.

m. Ability to write programs off-line.

4. Motor Control Modes

a. The drive shall be supplied with a Start-up and Auto-tune mode.

b. The V/Hz mode shall be programmable for fan curve or full custom patterns.

c. Capable of Open Loop V/Hz.

5. Current Limit

a. Programmable current limit from 20 to 160% of rated output current.

b. Current limit shall be active for all drive states: accelerating, constant speed and decelerating.

c. The drive shall employ PI regulation with an adjustable gain for smooth transition in and out of current limit.

6. Acceleration / Deceleration

a. Accel/Decel settings shall provide separate adjustments to allow either setting to be adjusted from 0 to 3600 seconds.

b. A second set of remotely selectable accel/decel settings shall be accessible through digital inputs.

7. Speed Profiles

a. Programming capability shall allow the user to produce speed profiles with linear acceleration/deceleration or "S Curve" profiles that provide changing accel/decel rates.

b. S Curve profiles shall be adjustable.

8. Adjustments

a. A digital interface can be used for all set-up, operation and adjustment settings.

b. All adjustments shall be stored in nonvolatile memory.

c. No potentiometer adjustments shall be required.

d. Nonvolatile memory for factory default values shall be provided.

e. Software must be available for trending and diagnostics, as well as online and offline programming functionality.

9. Process PID Control

a. The drive shall incorporate an internal process PI regulator with proportional and integral gain adjustments as well as error inversion and output clamping functions.

b. The feedback shall be configurable for normal or square root functions. If the feedback indicates that the process is moving away from the set-point, the regulator shall adjust the drive output until the feedback equals the reference.

c. Process control shall be capable of being enabled or disabled with a hardwire input. Transitioning in and out of process control shall be capable of being tuned for faster response by preloading the integrator.

d. Protection shall be provided for a loss of feedback or reference signal.

10. Skip Frequencies

a. Three adjustable set points that lock out continuous operation at frequencies which may produce mechanical resonance shall be provided.

b. The set points shall have a bandwidth adjustable from Maximum Reverse Speed to Maximum Forward Speed.

11. Fault Reset / Run

a. The drive shall provide up to nine automatic fault reset and restarts following a fault condition before locking out and requiring manual restart.

b. The automatic mode shall not be applicable to a ground fault, shorted output faults and other internal microprocessor faults.

c. The time between restarts shall be adjustable from 0.5 seconds to 30 seconds.

12. Run or Power Up

a. A user programmable restart function shall be provided to allow restart of the equipment after restoration of power after long duration power outages. Restart time dependent on presence of incoming signal.

13. Fault Memory

a. The last 32 fault codes shall be stored and time stamped in a fault buffer.

b. Information about the drive's condition at the time of the last fault such as operating frequency, output current, dc bus voltage and twenty-seven other status conditions shall be stored.

c. A power-up marker shall be provided at each power-up time to aid in analyzing fault data.

d. The last 32 alarm codes shall be stored and time stamped for additional troubleshooting reference.

14. Overload Protection

a. The drive shall provide internal class 10 adjustable overload protection.

b. Overload protection shall be speed sensitive and adjustable.

c. A viewable parameter shall store the overload usage.

15. Auto Economizer

a. An auto economizer feature shall be available to automatically reduce the output voltage when the drive is operating in an idle mode (drive output current less than programmed motor FLA). The voltage shall be reduced to minimize flux current in a lightly loaded motor thus reducing kW usage.

b. When the load increases, the drive shall automatically return to normal operation.

16. Terminal Blocks

a. Terminal blocks shall be provided for control wiring on all frames.

b. I/O terminal blocks shall be removable with wiring in place.

17. Flying Start

a. The drive shall be capable of determining the speed and direction of a spinning motor and adjust its output to "pick-up" the motor at the rotating speed. This feature is disabled by default.

18. Inputs and Outputs

a. The Input / Output option modules shall consist of both analog and digital I/O.

b. No jumpers or switches shall be required to configure digital inputs and outputs.

c. All digital input and output functions shall be fully programmable.

d. The control terminal blocks shall be rated for 115V AC.

e. Inputs shall be optically isolated from the drive control logic.

f. The control interface card shall provide input terminals for access to fixed drive functions that include start, stop, external fault, speed, and enable.

g. The VFD shall be capable of supporting up to 10 analog inputs, 10 analog outputs, 31 digital inputs, 10 relay outputs, 10 transistor outputs, and 5 positive temperature coefficient (PTC) inputs.

h. The Input / Output option modules shall have the following features:

i. Analog Inputs:

a. Quantity one (1) or two (2) differentially isolated, ±10V (bi-polar), 88k ohm input impedance, 11 bit plus sign.

b. Analog inputs shall be user programmable for a variety of uses including frequency command and process loop input. Analog inputs shall be user programmable for function scaling (including invert), offset, signal loss detect and square root.

ii. Analog Outputs:

a. Quantity one (1) or two (2) ±10V (bi-polar) / 11 bit & sign, 2 kΩ minimum load, 4-20 mA, 11 bit plus sign, 400 Ω maximum load.

b. The analog output shall be user programmable to be proportional to one of fourteen process parameters including output frequency, output current, encoder feedback, output power.

c. Programming shall be available to select either absolute or signed values of these parameters.

iii. Digital Inputs:

a. Quantity three (3) or six (6) digital inputs rated 24V DC/115V AC.

b. All inputs shall be individually programmable for multiple functions including: Start, Run, Stop, Auxiliary Fault, Speed Select, Jog and Process PI functions.

c. A single safety input shall be a configurable option for a thermostat or PTC temperature sensor.

iv. Digital Outputs:

a. At a minimum one (1) relay output (N.O. or N.C.).

b. For 240V AC or 24V DC, N.O. contact output ratings shall be 2 amps max., general purpose (inductive)/resistive. N.C. contact output ratings shall be 2 amps max., resistive only.

c. Relays shall be programmable to multiple conditions including: Fault, Alarm, At Speed, Drive Ready and PI Excess Error.

d. Timers shall be available for each output to control the amount of time, after the occurring event, that the output relay actually changes state.

e. Transistor outputs are available in quantities varying from zero (0) to two (2).

f. For 24V DC, transistor output rating shall be 1 amp max, Resistive.

19. Reference Signals

a. The drive shall be capable of using the following input reference signals:

i. Analog inputs

ii. Preset speeds

iii. Remote potentiometer

iv. Digital MOP

v. Human Interface Module

vi. Communication networks

20. Loss of Reference

a. The drive shall be capable of sensing reference loss conditions.

b. In the event of loss of the reference signal, the drive shall be user programmable to the following.

i. Fault the drive and coast to stop.

ii. Issue a minor fault - allows the drive to continue running while some types of faults are present.

iii. Alarm and maintain last reference.

c. When using a communications network to control the drive, the communications adapter shall have these configurable responses to network disruptions and controller idle (fault or program) conditions:

i. Fault

ii. Stop

iii. Zero Data

iv. Hold Last State

v. Send Fault Configuration

21. Metering

a. At a minimum, the following parameters shall be accessible through the Human Interface Module, if installed:

i. Output Current in Amps

ii. Output Voltage in Volts

iii. Output Power in kW

iv. Elapsed MWh

v. DC Bus Voltage

vi. Frequency

vii. Heatsink Temperature

viii. Last eight (32) faults

ix. Elapsed Run Time

x. IGBT Temperature

22. Faults

a. At a minimum, the following faults shall be accessible through the Human Interface Module:

i. Power Loss

ii. Undervoltage

iii. Overvoltage

iv. Motor Overload

v. Heat Sink Over-Temperature

vi. Maximum Retries

vii. Phase to Phase and Phase to Ground Faults

23. Predictive Diagnostics

a. At a minimum, the following predictive diagnostic features shall be provided:

i. Relay Output Life Cycles based on load type and amps.

ii. Hours of Fan Life based on load and ambient temperature.

iii. Motor Bearing life based on expected hours of use.

iv. Motor Lubrication schedule based on hours of use.

v. Machine Bearing life based on expected hours of use.

24. Real-Time Clock

a. Shall be capable of providing time stamped events.

b. Shall have the ability to be set locally or via a remote controller.

c. Shall provide the ability to be programmable for month, day, year and local time zones in HH:MM:SS.

2.4 VFD PACKAGED SYSTEMS

A. Features

1. Ratings

a. Voltage

i. Capable of accepting nominal plant power of 400, 480, 600 or 690V AC at 50Hz or 60Hz.

ii. The supply input voltage tolerance shall be ± 10% of nominal line voltage.

b. Displacement Power Factor

i. VFD shall be capable of maintaining a minimum true power factor (Displacement P.F. X Distortion P.F.) of 0.95 or better at rated load and nominal line voltage, over the entire speed range.

c. Efficiency

i. A minimum of 96.5% (+/- 1%) at 100% speed and 100% motor load at nominal line voltage.

ii. Control power supplies, control circuits, and cooling fans shall be included in all loss calculations.

d. Operating ambient temperature range without derating: 0 °C to 40 °C (32 °F to 104 °F).

e. Operating relative humidity range shall be 5% to 95% non-condensing.

f. Operating elevation shall be up to 4,500 ft. without derating.

2. Sizing

a. Systems rated at Normal Duty loads shall provide 110% overload capability for up to one minute and 150% for up to 3 seconds.

b. Systems rated at Heavy Duty loads shall provide 150% overload capability for up to one minute and 180% for up to 3 seconds.

c. Systems rated at Light Duty loads (Frames 8 to 10) shall provide 110% overload capability for up to one minute with no 3 second overload.

3. Auto Reset / Run

a. For faults other than those caused by a loss of power or any other non-critical fault, the drive system shall provide a means to automatically clear the fault and resume operation.

4. Ride-Through

a. The VFD system shall ride through power dips up to 15% of nominal. The duration of ride-through shall be inversely proportional to load. For outages greater than 15%, the drive shall stop the motor and issue a power loss alarm signal to a process controller, which may be forwarded to an external alarm signaling device. Provide self-reset capacity, and generator transfer wait/reset to restart motor when power is present at/above 15% threshold.

5. Run or Power Up

a. The VFD system shall provide circuitry to allow for remote restart of equipment after a power outage. Unless indicated in the contact drawings, faults due to power outages shall be remotely resettable. The VFD system shall indicate a loss of power to a process controller, which may be forwarded to an external alarm signaling device. Upon indication of power restoration, the process controller will attempt to clear any faults and issue a run command, if desired.

6. Communications

a. VFD shall provide an embedded Ethernet/IP port.

b. VFD shall be capable of communicating on multiple networks.

c. VFD shall be capable of supporting the protocol utilized for control, see control drawings.

7. Enclosure Door Mounted Human Interface Module (HIM)

a. VFD shall provide a HIM with integral LCD display, operating keys and programming keys.

b. An enclosure door-mounted HIM, rated NEMA/UL Type 1.

c. An optional VFD-mounted HIM, rated NEMA/UL Type 1, may be provided and shall be capable of connecting via a separate cable for use as a handheld terminal.

d. The HIM shall have the following features:

i. A seven (7) line by twenty-one (21) character backlit LCD display with graphics capability.

ii. Shall indicate drive operating conditions, adjustments and fault indications.

iii. Shall be configured to display in the following three distinct zones.

a. The top zone shall display the status of direction, drive condition, fault / alarm conditions and Auto / Manual mode.

b. The middle zone shall display drive output frequency.

c. The bottom zone shall be configurable as a display for either programming menus / information or as a two-line user display for two additional values utilizing scaled units.

iv. Shall provide digital speed control.

v. The keypad shall include programming keys, drive operating keys (Start, Stop, Direction, Jog and Speed Control), and numeric keys for direct entry.

B. Enclosure

1. Shall be rated NEMA/UL Type (1) or Type (12).

2. Shall be painted per the manufacturer's standard.

3. Shall provide entry and exit locations for power cables.

4. Shall contain a label for UL508.

5. The drive system nameplate shall be marked with system Short Circuit Current Rating (SCCR) meeting or exceeding available fault current shown on Drawings.

C. Drive Enclosure Input Disconnect

1. Provide an enclosure door interlocked disconnect.

2. Operator Handles:

a. Provide externally operated main disconnect handle.

b. Handles shall be lockable with up to three lockout / tagout padlock positions.

D. Control Power Transformer

1. Provide a control power transformer mounted and wired inside of the drive system enclosure.

2. The transformer shall be rated for the VFD power requirements.

E. Harmonic Mitigation

1. Drive Input Line Reactor

a. Provide a drive input line reactor mounted within the drive system enclosure.

b. The line reactor shall meet the following specifications.

i. The construction shall be iron core with an impedance of 5 percent.

ii. The winding shall be copper or aluminum wound.

iii. The insulation shall be Class H with a 115 °C rise over 50 °C ambient.

iv. The unit shall be rated for system voltage, ampacity, and frequency.

F. Auxiliary Relays

1. Provide relays for Drive Alarm, Drive Fault, Drive Run, and System Status Faults (as required).

2. The relay contacts shall be rated for 115V AC/30V DC, 5.0 amp resistive, 2.5 amp inductive.

G. Control Interface

1. The control terminals shall be rated for 115V AC.

2. The control interface shall provide input terminals for access to VFD functions that include start, stop, external fault, speed select, and enable, as required.

H. Hand/Off/Auto Selector Switch

1. Provide a "Hand/Off/Auto" selector switch, mounted on the enclosure door.

2. The "Hand/Off/Auto" selector switch shall start the drive in the “Hand” mode and stop the drive in the “Off” mode.

3. In the “Auto” mode the drive shall be started and stopped from a remote “RUN” contact.

4. In all modes, Auxiliary and Enable inputs to the drive control interface board must be present before the drive will start.

5. When a HIM is present, the stop function shall always be available to stop the drive regardless of the selected mode (“Hand” or “Auto”). The HIM will be non-functional (except for the display and programming) when the switch is in “Off” mode. The HIM shall stop the drive if the switch is in the “Auto” mode with the remote start contact initiated.

6. The drive speed reference shall be controlled from the HIM, unless a separate door-mounted potentiometer is provided, when in “Hand” mode (factory default setting).

7. The drive shall have the capability of smoothly transferring from the automatic speed reference to the manual speed reference on the HIM, without perturbation in the speed reference.

8. The drive speed reference shall be controlled by a remote 4…20 mA input when in “Auto” mode.

I. Drive Disable Mushroom Push Button

1. Provide a maintained mushroom style push button, mounted on the enclosure door that when pushed, will open the drive enable input.

J. Pilot Lights

1. Provide LED pilot lights, mounted on the enclosure door, for indication of the following status:

a. Run

b. Drive Fault

c. Control Power On

d. Motor Fault

K. Motor Run Time Meter

1. Provide a digital, non-resettable, door-mounted elapsed time meter.

2. The meter shall be electrically interlocked with the Drive Run relay and Bypass contactor to indicate actual motor operating hours.

L. Output Filtering VFD Shall include:

1. 5% output line reactor (Wells 1 & 3) and DV/DT output filter (Well 3).

2. No Voltage spikes at 100 HP submersible pump depth (greater than 200 feet length).

PART 3 - EXECUTION

3.1 EXAMINATION

A. Verify that location is ready to receive equipment.

B. Verify that the building environment can be maintained within the service conditions required by the manufacturer of the VFD.

3.2 INSTALLATION

A. Installation shall be in compliance with all manufacturer requirements, instructions and drawings.

3.3 START-UP SERVICES

A. At a minimum, the start-up service shall include:

1. Perform pre-Power Check
2. Megger Motor Resistances: Phase-to-Phase and Phase-to-Ground
3. Verify system grounding per manufacturer's specifications
4. Verify power and signal grounds
5. Check connections
6. Check environment

B. Drive Power-up and Commissioning:

1. Measure Incoming Power Phase-to-Phase and Phase-to-Ground
2. Measure DC Bus Voltage
3. Measure AC Current Unloaded and Loaded
4. Measure Output Voltage Phase-to-Phase and Phase-to-Ground
5. Verify input reference signal

C. All measurements shall be recorded.

D. Drive shall be tuned for system operation.

E. Drive parameter listing shall be provided.

END OF SECTION

SECTION 26 32 00
GENERATOR POWER SYSTEM

PART 1 GENERAL

1.1 REFERENCE STANDARDS

A. NFPA: 70, 76A

B. MIL

C. UL

1.2 DESCRIPTION

A. The Contractor shall provide all incidentals and materials and labor and install a complete and operable power system. Provide number of generators shown on plans.

B. This Specification covers requirements for providing a new, factory built, factory tested, field proven complete and operable standby diesel engine generators systems, including all equipment specified herein, shown on the drawings, and/or required for the service.

C. The standby system is to include an automatic transfer switches, engines/generators controls, batteries and other components, accessories, tests, documents and services as needed to meet the performance requirements of this specification and to provide a complete and operational system.

1.3 RESPONSIBILITY

A. This equipment will be manufactured by a single manufacturer who has been regularly engaged in the production of engine/generator sets, generators, transfer switches and controls for a minimum of ten years, so as to provide single source responsibility to the owner for warranty, parts and service. The manufacturer shall have local representatives who can provide factory trained service personnel, an adequate stock of replacement parts, and technical assistance for the entire standby system.

1.4 WARRANTY

A. Five year or 1800 hours of operation, whichever occurs first, from the time of substantial completion. The entire warranty shall be supplied by one manufacturer with technical expertise and adequate resources to provide adequate warranty support.

1.5 SUBMITTALS

A. Provide shop drawings per Requirements.

1.6 O & M MANUAL

A. Provide complete operation maintenance manuals and parts catalogues for all items comprising the standby system per requirements.

1.7 FACTORY TESTING

A. Testing: The intent of this specification is to provide equipment of proven reliability and compatibility. Factory prototype model tests, factory production model tests and field tests shall be performed and certified.

B. Prototype testing shall be performed on separate prototype models, not on the equipment sold or installed. Certification shall be submitted certifying the following tests have been performed.

1. Maximum power level (maximum KW).

2. Maximum motor starting capacity (maximum KVA).

3. Structural soundness.

4. Torsional analysis per MIL-STD 705B, Method 504.2 (Separate calculations based on engine or generator are not acceptable).

5. Engine-alternator cooling air flow.

6. Alternator temperature rise.

7. Harmonic analysis and voltage wave form deviation per MIL-STD-705B, method 601.4.

8. Three-phase short circuit test for mechanical and electrical strength.

9. Failure mode test for voltage regulator.

10. Endurance test at rated load and speed.

C. Factory production model tests shall be made, prior to shipment, of all system components. These tests shall be performed under rated load and power factor (unity power factor is not acceptable). Other tests shall include:

1. Single step load pickup per NFPA 76A.

2. Transient response and steady state governing.

3. Functional compatibility between generator set controls and transfer switches controls.

PART 2 PRODUCTS

2.1 ENGINE GENERATORS SETS

A. This systems shall include engines generators sets rated 0.8 power factor, 60 Hz, 3 phase, 4 wire, 480/277V on a continuous standby basis sized as noted on One-Line. The generators sets shall be mounted on a steel base suitable for mounting on a level surface. Vibration isolators are to be provided between engine generator and skid.

B. The engine shall be stationary, radiator cooled, diesel for use with Number 2 diesel fuel. It shall be four-cycle, with a minimum: Four (4) cylinders, 504 cubic inch displacement, capable of producing 192 BHP at rated speed. Provide DEF tank as required. Provide DEF continuous level indicators (low-DEF fluid only indicator not acceptable).

C. The engine equipment shall include the following:

1. Remote two wire negative ground starting system. Positive shift, gear engaging starter. With two independent methods provided to disconnect the starting circuit upon engine starting. The starting system shall be either twelve (12) or twenty-four (24) VDC.

2. Positive displacement, mechanical full pressure lubrication pump, oil filters, oil level indicator, and oil drain valve with hose extension.

3. Primary and secondary fuel filters with replaceable elements, electric fuel transfer pump, automatic fuel shut-off, replaceable dry element air cleaner, all mounted on engine.

4. Gear driven mechanical governor, with adjustable speed regulation of 5% from no load to rated load, automatic overspeed shutdown.

5. Low coolant level shut-down and high engine temperature shut-down.

6. Water temperature gauge, oil pressure gauge battery charger ammeter.

7. Engine mounted water jacket heater, 480/277 VAC, 1 phase, 1500 watt. OK per voltage shown. Means of disconnect provided when engine is running.

2.2 ENGINE COOLING SYSTEM

A. Shall be radiator cooled with belt or direct driven pusher fan, coolant pump, thermostat coolant resistant. Radiator shall have duct adapter flange for cooling duct.

B. Coolant system shall be filled with 50% Ethylene-glycol antifreeze/water solution as recommended by manufacturer.

2.3 ENGINE EXHAUST SYSTEM

A. Residential grade silencer with condensate trap. Sized as recommended by engine manufacturer.

B. Stainless steel flexible exhaust connection.

2.4 FUEL SYSTEM

A. Double shell full size skid tank with overflow/leak alarm sized not less 24 hour required operation time at full load.

B. Integral fuel delivery system.

C. Venting as required with stack mounted away from combustibles.

D. Provide full tank after required testing, Commissioning, etc.

2.5 GENERATORS

A. Generators shall be single bearing, self-aligning, four pole, synchronous type, revolving field, with amortisseur windings, direct drive centrifugal blower for proper cooling, asynchronous solid state voltage regulator, with brushless rotating rectifier exciter system. No brushes allowed. Generators shall be direct connected to the engine flywheel housing, driven through a flexible coupling to ensure proper alignment. Gear driven generators are not acceptable. Class F insulation.

B. Three phase, broad range, twelve lead reconnectable generators.

C. Frequency regulation shall not exceed 3 Hz from no load to rated load. Voltage regulator shall be solid-state design and shall function by controlling the exciter magnetic field between stator and rotor to provide no load to full load regulation within +/- 1.0% of rated voltage during steady-state conditions. The generators sets and regulator must sustain at least 90% of no-load voltage for ten seconds with 250% of rated load at near zero power factor connected to its terminals. A rheostat or solid-state control shall provide a minimum of +/- 5.0% voltage adjustment from rated value.

D. The alternator, exciter, and voltage regulator shall be designed and manufactured by the generators set manufacturer so that the characteristics shall be matched to the torque curve of the prime mover. The design shall allow the prime mover to utilize its fullest power producing capacity (without exceeding it or over compensating) at speeds lower than rated, to provide the fastest possible system recovery from transient speed dips. System shall provide automatic voltage reduction if the load demand exceeds the engine capacity to prevent engine stalling and saturation of magnetic components. Permanent Magnet Generators (PMG) Excitation type system.

E. Exciter shall be three phase, full-wave, rectified, with heavy-duty silicon diodes mounted on the common rotor shaft and sized for maximum motor starting capability. System using three-wire solid state control elements (such as transistors or SCR's) rotating on the rotor is not acceptable.

F. For electronic loads a "Torque Match 2" regulator shall be included.

2.6 ENGINE- GENERATORS CONTROLS

A. Provide a lighted, unit mounted, control console, vibration isolated mounted, wired and tested by the generators manufacturer. Terminals shall be identified as to their function, or purpose. Control terminals in the generators control panel and automatic transfer switch shall be identical for ease of connection by the Contractor.

B. Control console shall include the following:

1. Engine controls and gauges.

2. Three position selector switch (Run-Stop-Remote).

3. Contacts for engine start and stop.

4. Engine monitor (solid state) with fault lights and external alarm terminals for over crank, overspeed, high coolant temperature, low oil pressure and low engine coolant temperature. Engine shutdown provided for over crank, overspeed, high coolant temperature and low oil pressure shutdown. Pre-alarms shall be provided for high coolant temperature and low oil pressure. Contacts shall be provided for remote annunciation of the above.

5. Provide an adjustable solid-state cycle cranker which shall disconnect the starting control after sixty seconds and a minimum of three cranking attempts.

6. Asynchronous solid state voltage regulator with voltage adjusting rheostat.

7. Manual reset field circuit breaker.

8. Running time meter, AC voltmeter (dual range - indicating all voltages), AC ammeter (dual range), voltmeter/ammeter phase selector switch with OFF position and frequency meters. AC meters shall be 2% accuracy.

9. Mounted output circuit breaker, size as indicated on the drawings with AUX. Contacts each.

10. Generators Breaker(s), trip alarm.

2.7 ACCESSORIES

A. Heavy duty lead acid batteries with battery rack, as recommended by the engine manufacturer.

B. Vibration isolators, spring type, sized as recommended by generators manufacturer, minimum of four.

C. For Well#3- Complete WP insulated, sound insulated, and lockable Exterior enclosure. Residential Site sound attenuation level. Color per Civil.

2.8 AUTOMATIC TRANSFER SWITCHES

A. The complete automatic load transfer control shall be designed, built, and tested by the generator manufacturer. The load transfer control shall be rated for continuous duty and for all classes of load. The automatic load transfer control shall be provided with three poles for normal standby service of 480/277 Volt, 60 Hz, 3-phase.

B. Operation: The Automatic Load Transfer Control shall sense complete loss of normal power on any phase and signal the standby generating sets to start after normal power failure. When the standby power attains proper voltage and frequency, the Load Transfer Control will automatically transfer the load to standby power. This operation shall occur within 10 seconds. When the normal power is restored, the Automatic Load Transfer Control shall sense this and retransfer the load from standby power to normal power, and signal the standby source to stop.

C. Rating and Performance:

1. The transfer switches shall be rated for all classes of load including inductive and noninductive load at 600 Volt. The transfer switch portion of the control shall be designed, built, and tested to close on an inrush current up to and including twenty (20) times the continuous rating of the switch without welding or excessive burning of the contacts. The transfer switch shall be capable of switching the load up to and including fifteen (15) times the continuous rating of the switch and capable of enduring six thousand (6000) cycles of operation, at rated current, at a rate of six (6) cycles per minute, without failure. One cycle shall consist of one complete opening and closure of both sets of contacts on an inrush current of ten (10) times the continuous rating of the switch.

2. Transfer switches shall be rated for amps as shown and have a withstand current of not less than 30,000 amps (RMS-symmetrical) and as shown.

2.9 CONSTRUCTION

A. A indoor, NEMA 1 enclosures with key locked doors shall house the transfer switches and control components. The indicating lamps and meters shall mount on the front to be visible without opening doors.

B. Voltage sensors and time delays shall be solid-state, plug-in devices. The control relays shall be dust covered, plug-in devices. These control accessories shall mount on a dead-front, swing-out, control accessory panel to avoid generators shock hazard while adjusting control functions, but will swing out exposing the wiring to facilitate servicing.

C. The transfer switches, with terminal lugs for either copper or aluminum wire, shall have individual heat resistant chambers enclosing solid silver cadmium oxide, double-break contacts. The transfer switches, with mechanical and electrical and interlocks to prevent simultaneously energizing both normal and standby service shall be mechanically held on line side with auxiliary contacts rated 6-amp, 120-volt AC; 3-amp, 240-volt, AC on line side and generators side of transfer switch.

2.10 ACCESSORIES

A. The following accessories shall be provided with the Automatic Transfer Switches.

1. A time delay to start standby power, adjustable from 0 - 10 seconds, shall be provided to prevent needless starting and stopping during periods of momentary voltage fluctuations from the normal power source.

2. A time delay to pick-up load, adjustable from .5 - 30 seconds, shall be provided to prevent the standby power from accepting load until proper voltage and frequency is reached.

3. A time delay to retransfer the load, adjustable from 0 - 30 minutes, shall be provided to delay retransfer to avoid short term normal power restoration.

4. A time delay to stop the engine-generators after the load has been transferred to normal power shall be provided. The time delay, adjustable from 1 to 15 minutes, shall permit engine to run unloaded to cool down before shutdown.

5. Automatic bypass to re-transfer the load from generating set to normal power source if the standby set should fail during the delay period.

6. Provide battery charger, SCR voltage regulated type with float and taper features; 6-amp at 24 V.D.C.; 10- amp at 12 V.D.C. as required for generators set. Charger shall have charging ammeter and fuse protection. Charger shall not be damaged during engine cranking/running.

   a. Integrated charger with control system is not acceptable. Separate battery charging unit required.

7. Provide a test switch to simulate an interruption of power from the normal source.

8. Provide an exerciser clock to automatically start the generating set at regular intervals and allow it to run for a preset time period.

9. A load selector switch shall be provided to test or exercise generators with or without building load.

10. Adjustable solid state voltage sensors, one per phase on line side. Sensors shall be temperature compensated over the temperature range of 225 degrees F. to +175 degrees F.

11. Disconnect plug to electrically disconnect the control section from the transfer switch for maintenance service during normal operation.

12. Normal and emergency indicating lamps to indicate which source is supplying power to the load shall be provided on the front of the enclosure door.

13. A.C. ammeter 0 - 6 amp for battery charger.

14. Over crank indicator lamp.

2.11 MANUFACTURER

A. Kohler 150REOZJF Series for Well #1 fitment, and airflow coordinated or approved.

B. Kohler, CAT, Onan for Well #3 fitment coordinated, or approved.

C. Well#1 and Well#3 genset manufacturers shall match.

D. Manual transfer switches (see drawings) shall be by same manufacturer as ATS's (Generators)

PART 3 - EXECUTION

3.1 INSTALLATION

A. Follow manufacturer's recommendations and requirements explicitly.

B. Engine generators set shall be installed as required when shown on the drawings.

C. The Electrical Contractor shall fill the fuel oil storage tank with No. 2 diesel fuel for testing and final checking prior to acceptance. Provide fuel to completely fill the tank to capacity.

D. Provide and mount on level elevated concrete pad outdoors, away from peril. Bolt to pad.

3.2 START-UP

A. Factory Start-up: A qualified factory technician will:

1. Certify installation.

2. Check installation and interface connections to insure proper mating of generators system with building wiring.

3. Notify Owner and Engineer 72 hours in advance prior to Operation Test.

4. Perform on-site load testing which includes:

   a. Two hours at 50% of generators set rating (can be performed at factory).

   b. Simulated power failure tests utilizing building load and using the transfer switch and its time delays (two hours).

   c. A fifteen-minute engine/generators cool down period without load before shutdown.

   d. All testing shall include records at fifteen-minute intervals of water temperature, oil pressure, ambient air temperature, voltage, current, frequency, KW and power factor. These tests shall be forwarded, in triplicate, to the Owner.

5. Fill engine cooling system with ethylene-glycol antifreeze and water solution.

3.3 GENERATORS TIME SETTINGS

A. Loss of normal power to power available: within 10 seconds.

B. Load re-transfer back to utility: 15 minutes.

C. Cool down period (no load): 10 minutes, or as recommended by generators manufacturer.

END OF SECTION

SECTION 27 05 00
DATA SYSTEM

PART 1 - GENERAL

1.1 WORK INCLUDED

A. Provide Data system infrastructure including: cables, racks, support, hardware, and wall plates for complete functional data communications as shown on Drawings and specified herein.

1.2 SUBMITTALS

A. See Execution herein. Provide Shop Drawing showing detailed diagram of system infrastructure within wellhouses.

1.3 APPLICABLE STANDARDS

A. Work shall be provided per Code. Provide per manufacturers requirements and recommendations. Materials shall be UL approved or listed by accredited and approved listing agency. Equipment shall be labeled where a UL standard and/or test exists.

B. The latest edition of referenced standards from the latest available draft in the case of proposed standards; shall be utilized.

1. Underwriters Laboratories (UL)

   a. UL 444 Communications Cables

   b. UL 1863 Standard for Communications-Circuit Accessories

   c. UL 2269 Optical Fiber/Communications/Signaling/Coaxial Cable Outlet Boxes

2. Underwriters Laboratories

   a. ULC-S139 Fire Test for Evaluation of Integrity of Electrical Cables

3. American National Standards Institute (ANSI), Telecommunications Industry Association (TIA), Electronic Industries Alliance (EIA)

   a. ANSI/TIA/EIA-492 AAA Multi Mode Optical Fiber

   b. ANSI/TIA/EIA-568-A Commercial Building Wiring Standard

   c. ANSI/TIA/EIA-568-B Commercial Building Telecommunications Cabling Standard

   d. ANSI/TIA/EIA-854 A Full Duplex Ethernet Specification for 1000Mbis/s (1000BASE-TX) Operation Over Category 6 Balanced Twisted-Pair Cabling.

e. ANSI/TIA/EIA 607 Commercial Building Grounding and Bonding Requirements for Telecommunications.

4. International Standards Organization (ISO), International Electrotechnical Commission (IEC)

a. ISO/IEC 11801 Generic Cabling for Customer Premises

b. ISO/IEC 60603-7 RJ45 Connector (8-Pin)

c. ISO/IEC 60793-2 Optical Fibers

5. Electrical Code

6. National Electrical Manufacturers Association (NEMA)

a. NEMA WC-63 Performance Standards for Twisted Pair Premise Voice and Data Communications Cable

b. NEMA WC- 66 Performance Standard for Category 6 and Category 7 100 Ohm Shielded and Unshielded Twisted Pair Cables

7. ANSI, Insulated Cable Engineers Association (ICEA)

a. ANSI/ICEA S-90-661 Category 3, 5, & 5e Individually Unshielded Twisted Pair Indoor Cable for Use In General Purpose and LAN Communication Wiring Systems.

C. Applicable Testing Standards: Testing of individual components and channel shall be conducted in accordance with the following standards:

1. ASTM D 4566-94, Standard Test Methods for Electrical Performance Properties of Insulation and Jackets for Telecommunications Wire and Cable

2. ANSI/TIA/EIA-492 AAA, Multi Mode Optical Fiber

3. ANSI/TIA/EIA-568-A, Commercial Building Telecommunications Standard

4. ANSI/TIA/EIA-568-A-1, Propagation Delay and Delay Skew Specifications for 100 W 4-pair cable

5. ANSI/TIA/EIA-568-A-2, Corrections and Additions to TIA/EIA-568-A

6. ANSI/TIA/EIA-568-A-4, Production Modular NEXT Loss Test Method and Requirements for Unshielded Twisted Pair Cabling

7. ANSI/TIA/EIA-568-A-5, Transmission Performance Specifications for 4-pair 100 W Category 5e Cabling

8. ANSI/TIA/EIA-TSB 67, Transmission Performance Specifications for Field Testing of Unshielded Twisted Pair Cabling Systems

9. ISO/IEC 11801

PART 2 - PRODUCTS

2.1 COPPER CABLE (BUILDING AND PATCH CORDS)

A. Category 6

B. Meets ULC S139

C. Copper twisted 4-pair, unshielded

D. Plenum rated where routed through plenums

E. Impregnated white or blue jacket.

F. Acceptable cable manufacturers: Avaya, LANmark, Beldon, Mohawk, Corning.

2.2 FIBER OPTIC CABLE

A. Outdoor purpose single-mode. 9 micron diameter, 125 micron cladding. Type SC Termination Connectors.

B. 5km distance, gel-filled/Outdoor rated (OSP) in raceway per drawings.

C. Multi-Pair 12 Strand

D. Meets ULC-S139

E. Bandwidth shall meet or exceed 1000BASE-LX

F. Fiber Cable: Corning, CommScope, or approved

PART 3 - EXECUTION

3.1 GENERAL

A. It is the intent of the Drawings and Specifications that systems be complete, fully tested and ready for operation.

B. Data systems are shown and presented as a performance specification. Submit Shop Drawings, including connection points, Risers, and the like for review. Specific suite and common area requirements have been indicated, Contractor shall provide detailed drawings showing cabling and riser information.

C. In general, the Project work includes installation of new horizontal station cable, jacks, face plates, outlets, distribution frame hardware, quantified patch cable, punch blocks, cable terminations, labeling, grounding and bonding, and conformance testing of the installation in addition to other requirements specified herein.

D. Contractor shall provide cable support hardware in all locations to vertical secondary pathway at outlet location. Utilize cable tray to the extent possible.

E. Bundle cable groups utilizing equipment specified. Provide re-usable straps and locate to the extent possible near accesses. Do not use Nylon Cable Ties. Do not staple cables.

F. The Structured Cabling System (SCS) shall follow ANSI/TIA/EIA and BICSI guidelines. For direct assistance in interpreting telecommunications standards, utilize the services of a Registered Communications Distribution Designer (RCDD) certified by the Building Industry Consulting Service International (BICSI).

G. Coordinate with the Owner for all labeling. Re-apply labeling that has not been coordinated or that is not permanent.

H. Route all cables in a neat a professional manor.

I. Secure all cable, loose or "sagging" cable across large spans is not acceptable.

J. Do not pinch or staple cables in any locations. Pinched or stapled cables are not acceptable.

K. Route cables and maintain bending radius of cables through raceway as required by cable manufacturer and applicable standards.

L. Do not splice cables.

3.2 IDENTIFICATION

A. Label all equipment as required by Specifications and as shown on drawings.

B. Coordinate with owner and verify labeling method used to identify areas within building. Regardless of Owner requirements, include labeling indicators as shown on Drawings.

3.3 GROUNDING

A. Ground all equipment racks utilizing ground bus bars and grounding conductor not less than as required by Code and standards.

B. Utilize compression lugs at all ground bus locations.

3.4 TESTING

A. Copper Cables

1. Provide testing for performance with IEEE Category 6 requirements, and ANSI/TIA/EIA performance requirements.

2. Visually inspect all cables and packaging to detect cable damage incurred during shipping and transport. Return visibly damaged items to the manufacturer.

3. Conduct cable testing as described below upon completion of installation. Test fully completed systems only. Sample or Piecemeal testing is not acceptable, except by prior written approval from the Architect and Owner.

4. After terminating both ends, but before any cross connects are installed, test all voice and data stations cables for attenuation and for near-end cross talk (NEXT) to CAT 6. Test all UTP backbone, distribution and patch cable for cable pair/conductor continuity, ground fault, proper cross-connect, shorts, loose connectors, and crossed pairs.

5. Remove all defective cables from pathways system. Do not abandon cables in place.

6. Maintain and complete a comprehensive typed testing log showing testing results of all cables. Log shall include cable identification as required herein and shown. Indicate cables that were replaced as a result of testing. Submit Log for Owner with O&M manuals. Execution of testing log does not alive Contractor from providing functional equipment as required herein regardless of testing results.

B. Fiber Cables

1. Visually inspect all fiber, fiber reels, and shipping cartons to detect cable damage incurred during shipping and transport. Replace visibly damaged items.
2. Conduct fiber testing as described below upon completion of installation. Test fully completed systems only. Piecemeal testing is not acceptable, except by prior written approval from the Engineer.

3. After installation of connectors, visually inspect each fiber end-face at 10x magnification. Refinish fibers with visible defects and/or variations in the core area.

4. Perform end-to-end, bi-directional attenuation (loss) test for each fiber strand at 850 nm and 1300nm wave lengths. Conduct tests in accordance with EIA/TIA-526-14, method B and with test instrument manufacturers printed instructions test equipment must have calibration certificate.

5. Demonstrate that measured link loss does not exceed the allowable loss, which is the sum of: the connectors loss (based on the number of mated connector pairs at the EIA/TIA-568 maximum allowable loss of 3.75 db/km @ 850nm and 1.5dB/km @ 1300nm).

6. Strands whose measured attenuation falls outside the acceptable range shall be subject to further inspection and testing to determine the nature of the fault. At a minimum, an OTDR shall be used to determine the true loss for each connector pair, the exact length of the fiber, and to identify the presence of any core damage.

7. Faults related to connectors shall be corrected, and the fiber re-tested as stated in the paragraph above until acceptable attenuation measurements are recorded.

8. Where defects are found to be inherent in the fiber itself, fiber must be removed and replaced. Also, replace any cable having fewer than the manufacturers guarantee number of serviceable fibers.

9. Remove all defective cables from pathways system. Do not abandon cables in place.

10. The Engineer reserves the right to observe the conduct of any or all portions of the testing process.

11. All test results and corrective procedures are to be documented and submitted to the Owner within fourteen (14) working days of test completion.

3.5 ACCEPTANCE

A. Upon receipt of the Contractor's log of cable testing, the Engineer will review the installation and may request a retest using Contractor equipment and labor, of up to 10% of the cable/wires installed. Inconsistent results are not acceptable.

END OF SECTION